Game Theory

〈 Evolutionary Games and Spatial Chaos : Spatial Prisoner's Dilemma 〉

Group 12

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Outline

- Introduction of Spatial Prisoner's Dilemma
- Spatial Prisoner's Dilemma Game
- Programming simulation
- Deep Learning Prisoners Dilemma Evaluation
- application
- conclusion

Introduction of Spatial Prisoner's Dilemma

- **Canonical Game Theory Problem:**
  Two players choose to Cooperate or Defect, with outcomes illustrating conflict between individual interest vs mutual benefit
- **Payoff Structure:**
  Mutual cooperation yields moderate benefit for both
- **Dilemma Nature:**
  Rational self-interest leads both to defect

Spatial Prisoner's Dilemma Game

- Two players: Cooperate (C) or Defect (D)
- Payoff matrix: $T > R > P \geq S$

	C'	D'
C	$R = 1$	$S = 0$
D	$T = B > 1$	$P = 0$

C: cooperate D:defect

- Rational choice typically leads to defection

Spatial Prisoner's Dilemma Game

- Two players: Cooperate (C) or Defect (D)

顏色(color)	狀態轉換(State Transition)
藍色(blue)	C → C
綠色(green)	C → D
黃色(yellow)	D → C
紅色(red)	D → D

Spatial Prisoner's Dilemma Game

- Players are placed on a 2D grid (e.g., 99x99).
- Each round:
- Play PD with 8 neighbors
- Replace each site with the highest scoring neighbor

fc value	**Outcomes**
fc continuously drops to 0	Cooperation is completely eliminated by defection (Collapse)
fc stabilizes at a certain value (e.g., 0.32)	Cooperation and defection reach stable spatial coexistence
fc oscillates within a certain range	Chaotic dynamic coexistence
fc rapidly rises to 1	Cooperation achieves total victory

Programming simulation

b=1.7;p=0.1;fc≈0.74

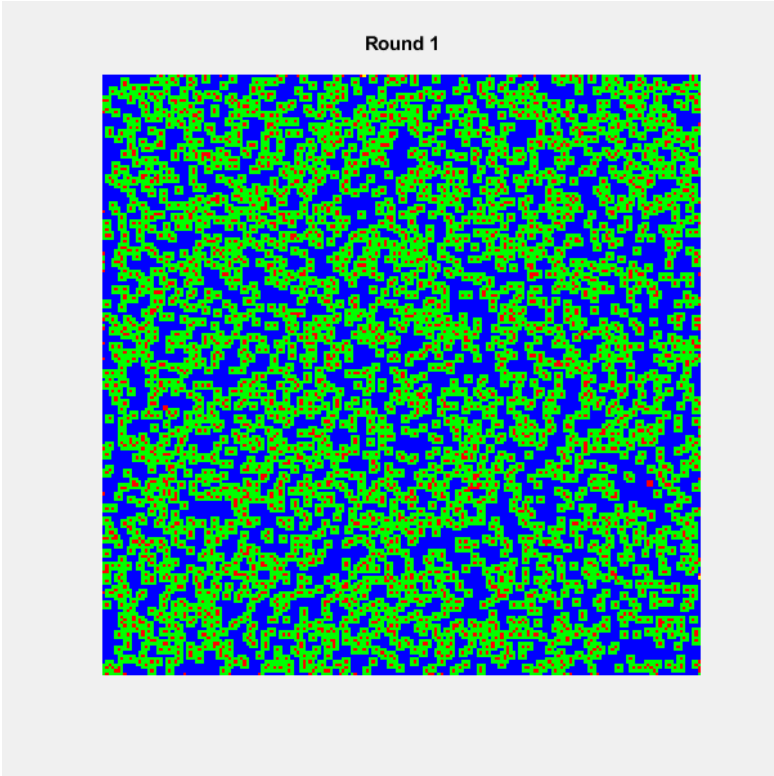


Fig1

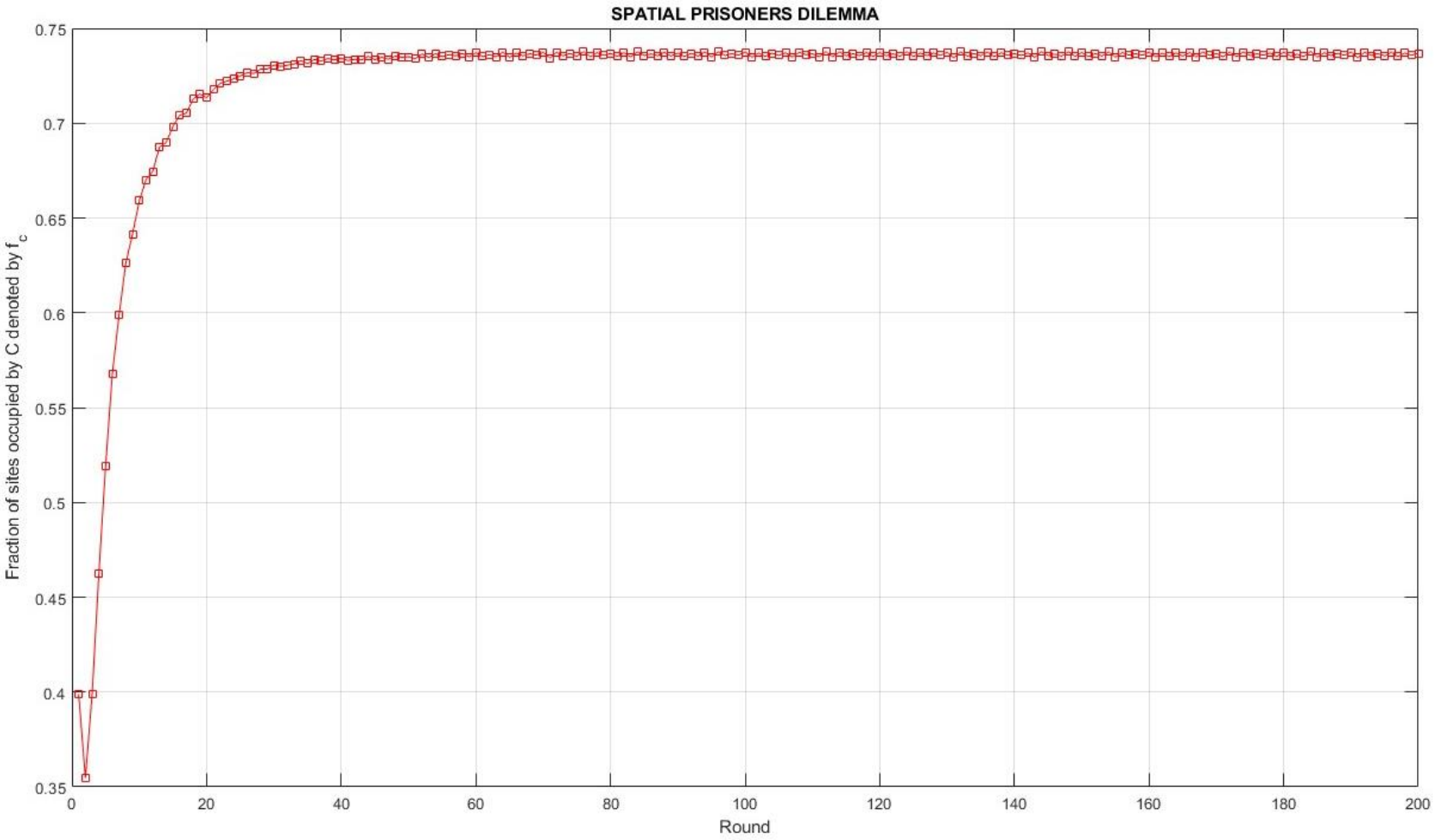


Fig1_fc

Programming simulation

b=1.90;p=0.1;fc≈0.318

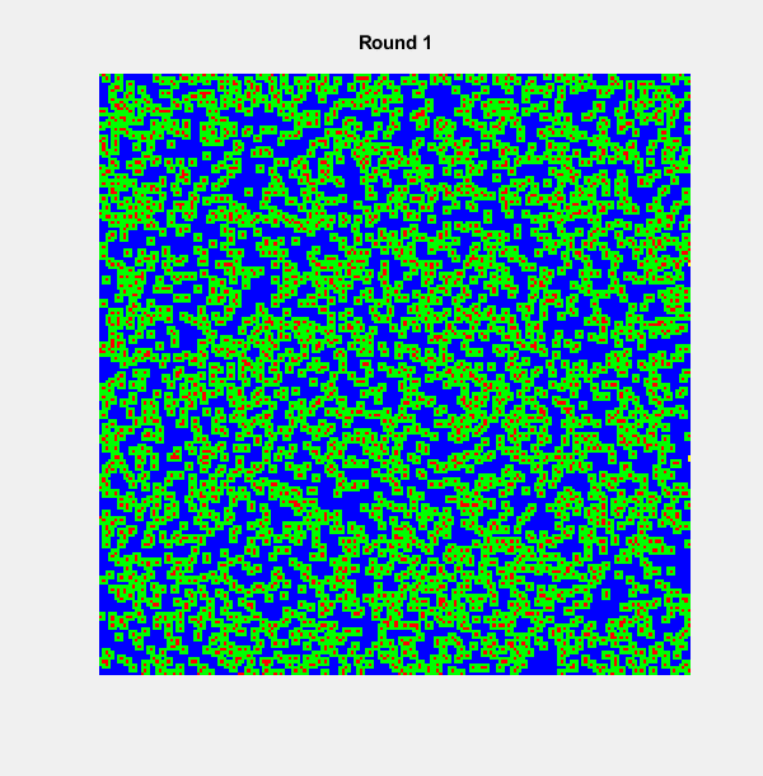


Fig2

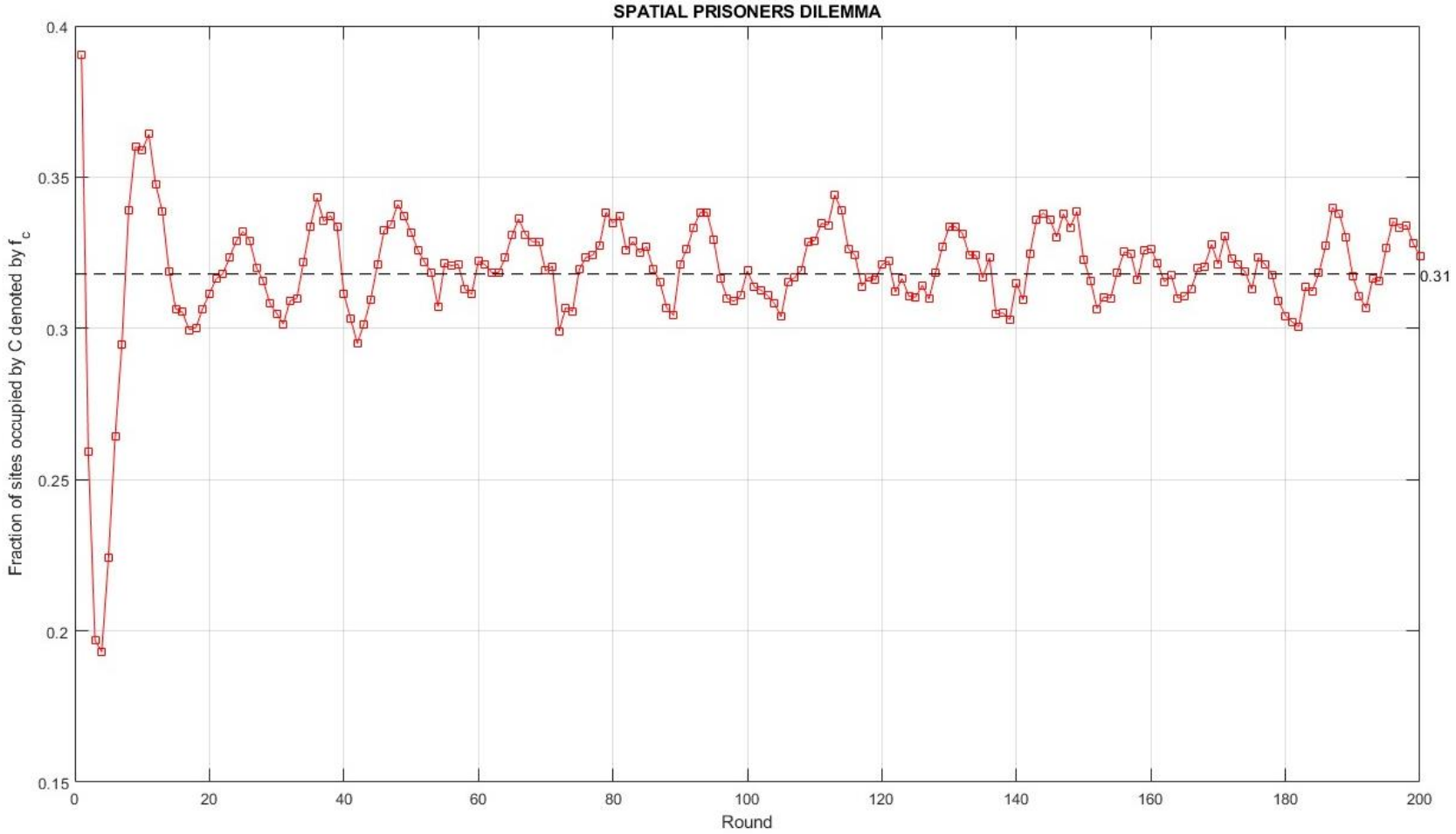


Fig2_fc

Programming simulation

b=2.50;p=0.1;fc≈0.0

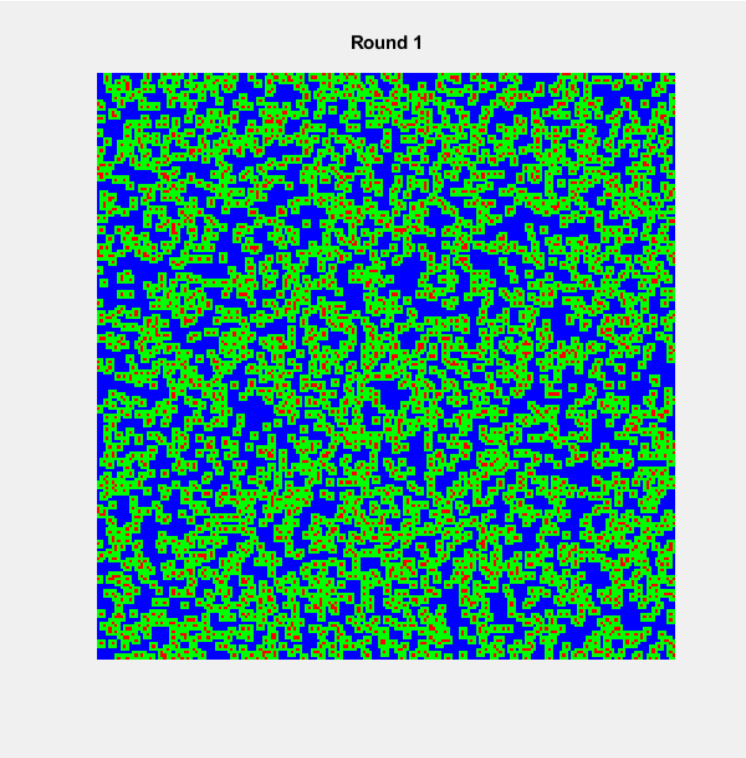


Fig3

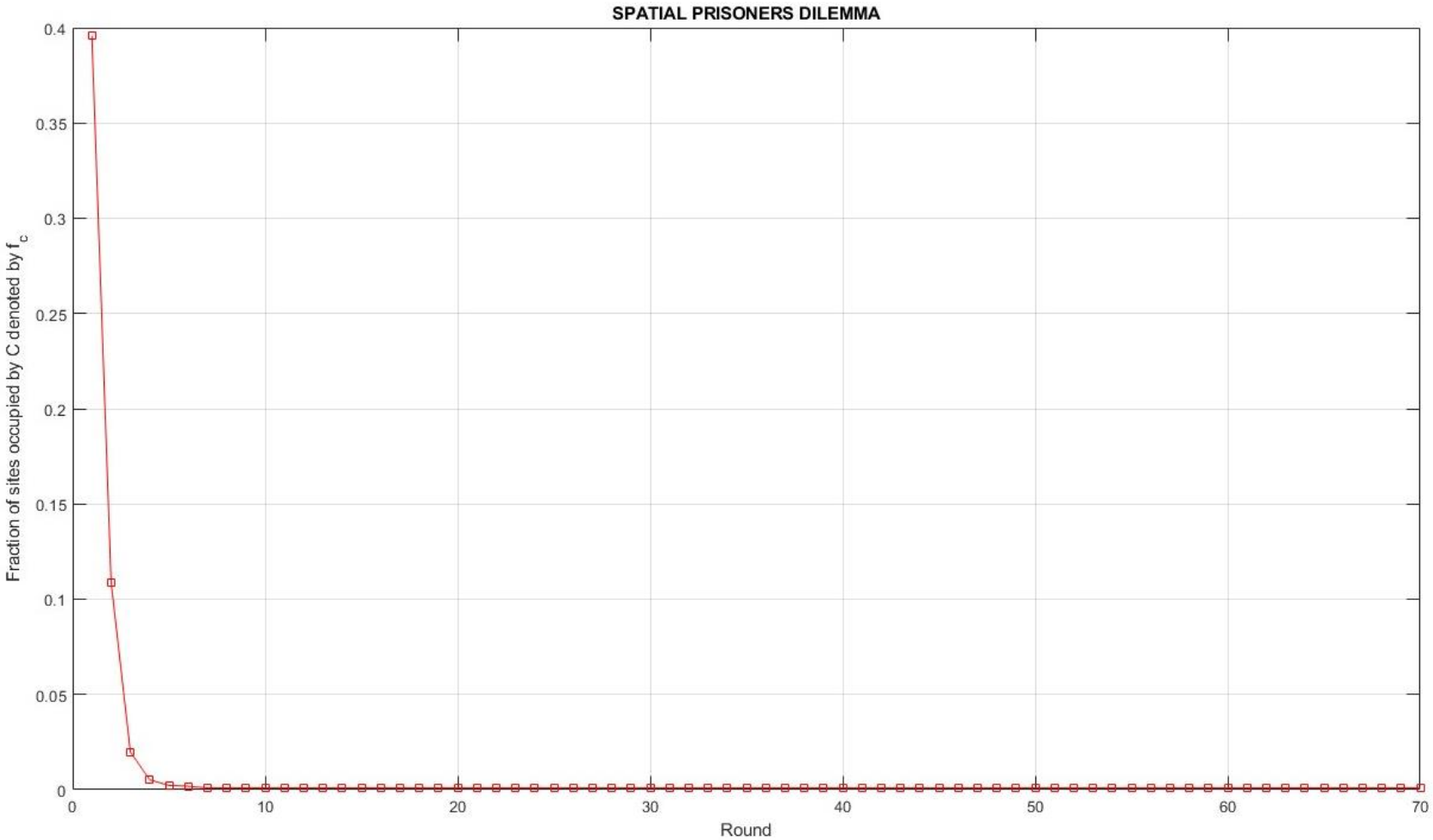


Fig3_fc

Programming simulation---discussion

b value	P value	Self interaction	fc	Outcome remarks
1.7	0.1	1	0.74	Cooperation dominates
1.90	0.1	1	0.32	Dynamic coexistence
2.5	0.1	1	0	Defection dominates

1. As 'b' increases, the payoff for defection tempts more players to become D.

2. The system transitions from stable cooperation → chaos → near-total defection, exhibiting a typical critical phenomenon of cooperation collapse.

Programming simulation

b=1.70 ; p=0.1;fc≈0.72

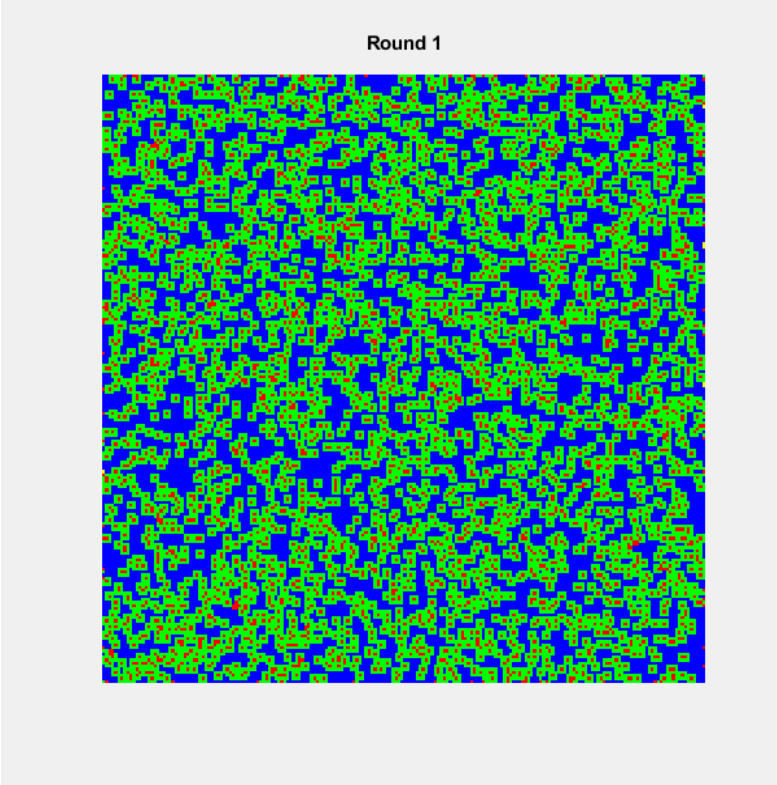


Fig6

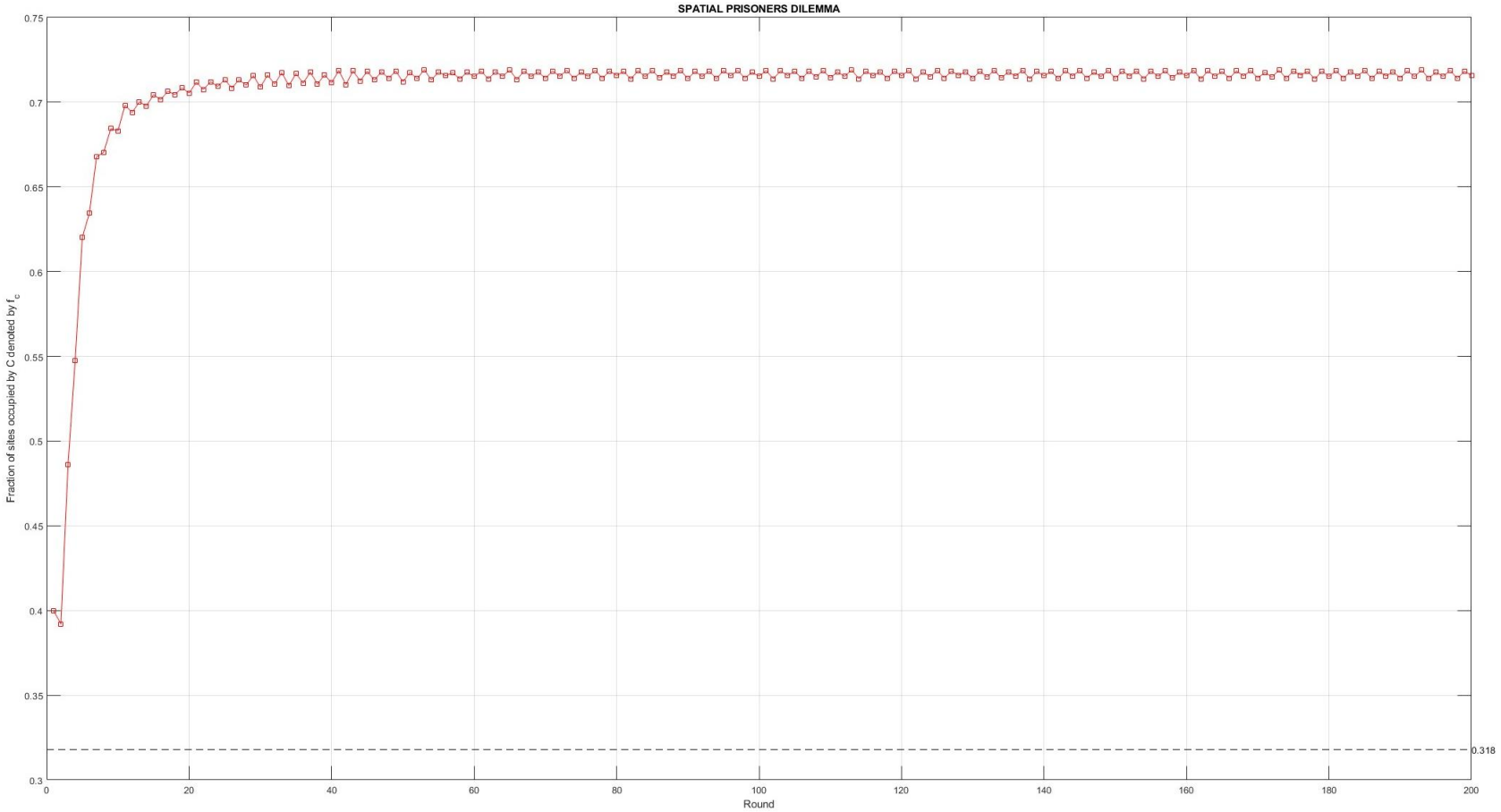


Fig6_fc

Programming simulation

b=1.70; P=0.6;fc≈0.91

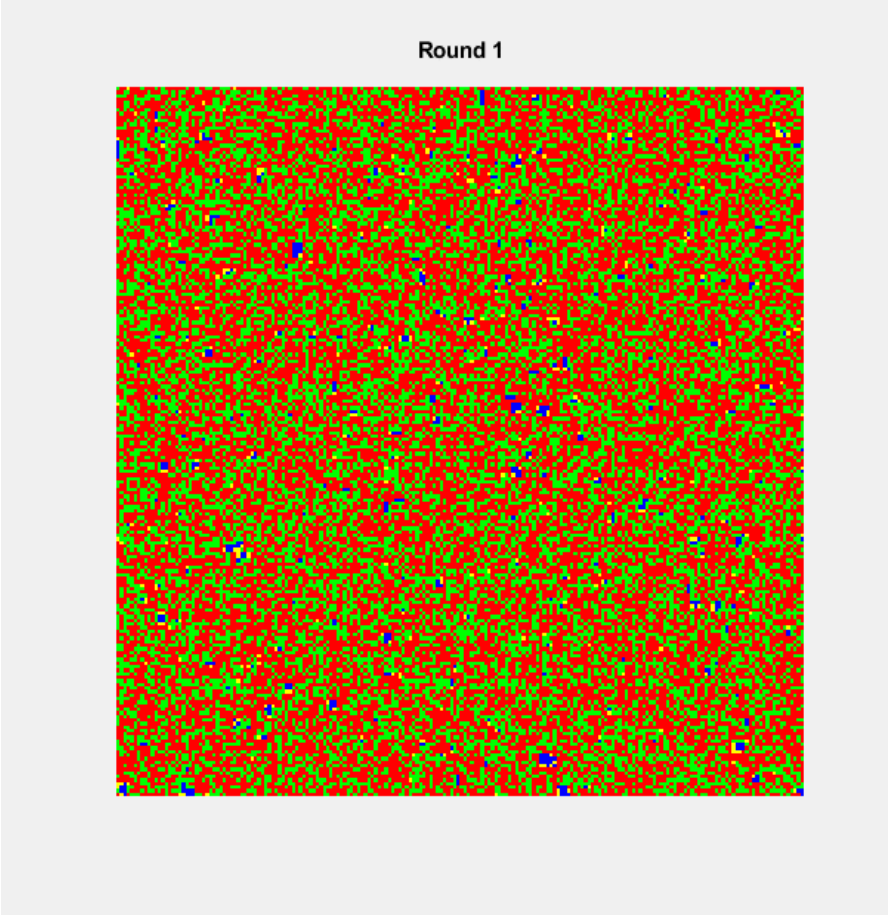


Fig7

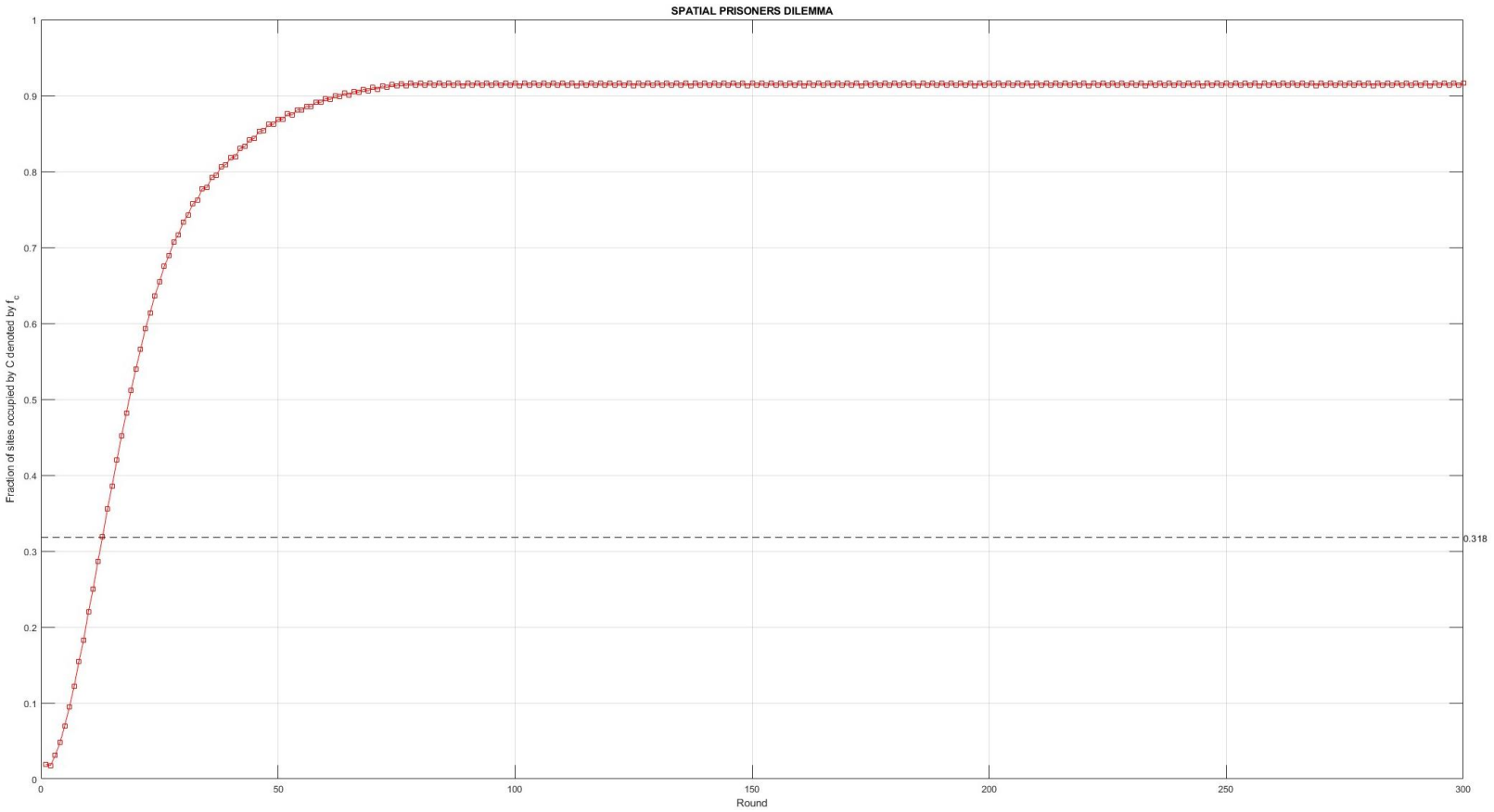


Fig7_fc

Programming simulation

b=1.90;p=0.1;fc≈0.318

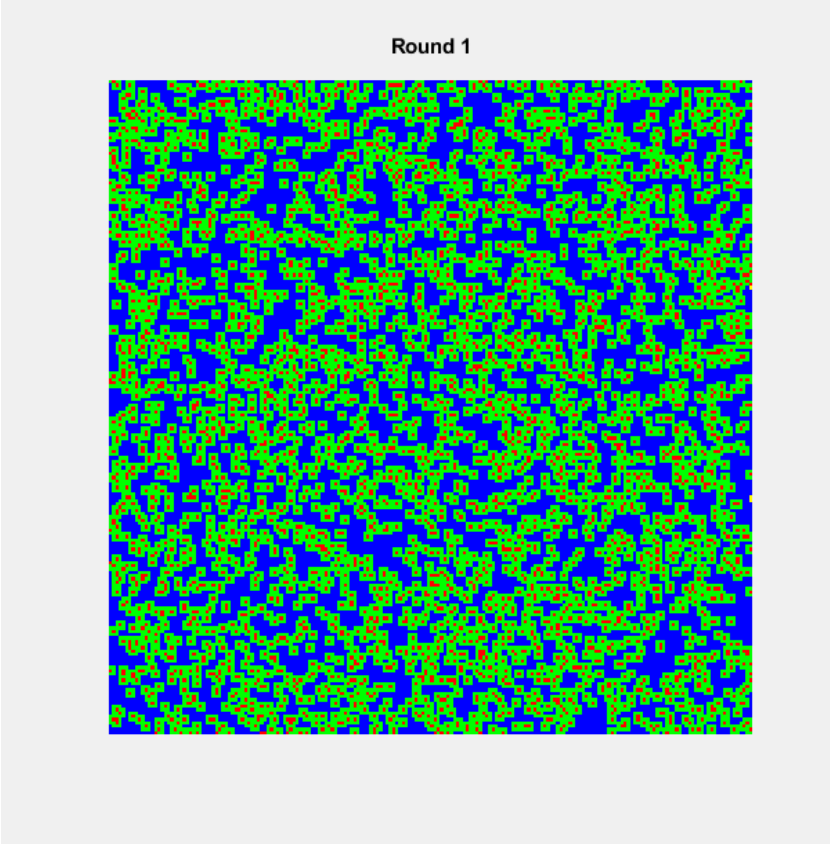


Fig4

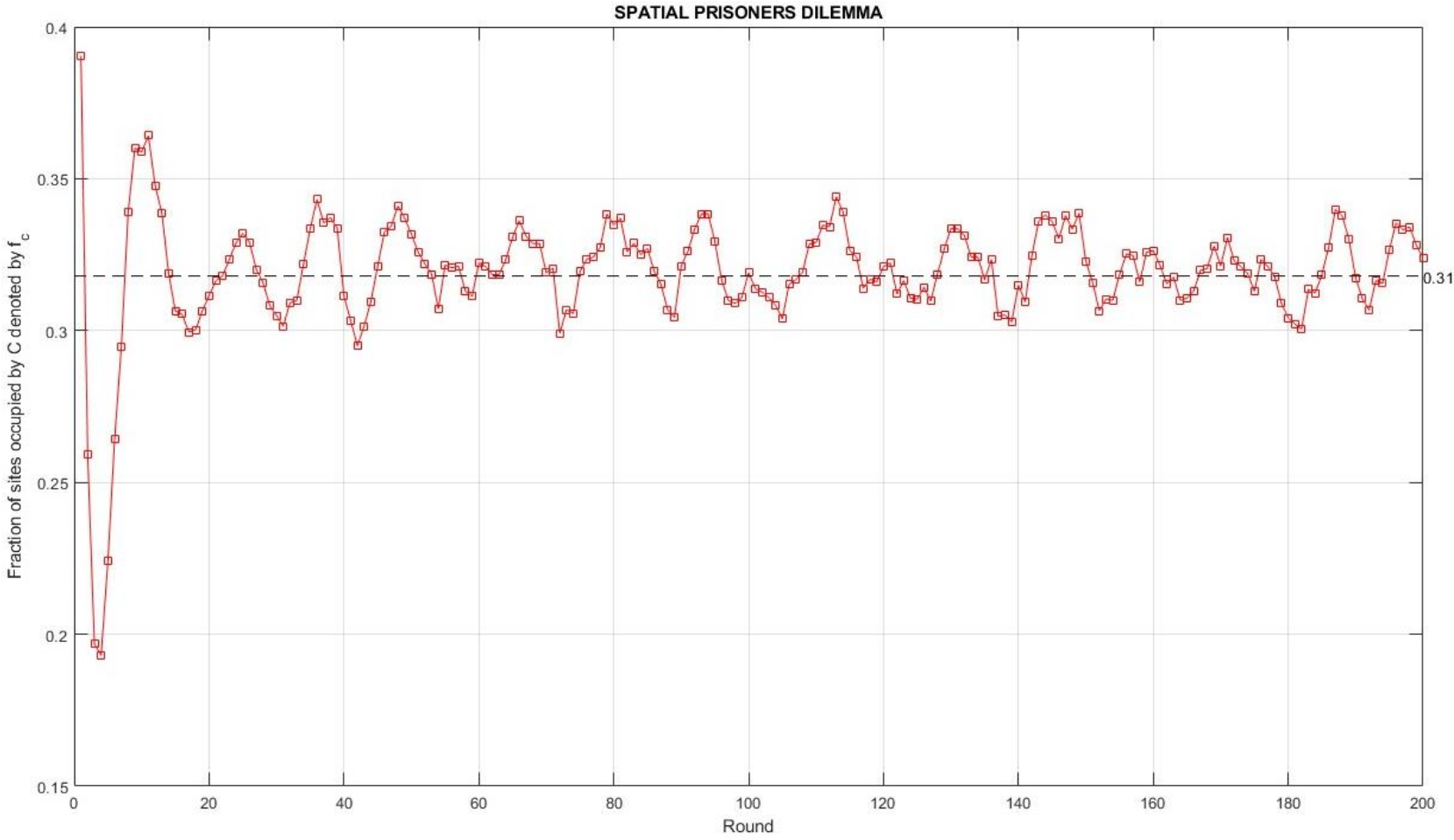


Fig4_fc

Programming simulation

b=1.90; P=0.6;fc≈0.318

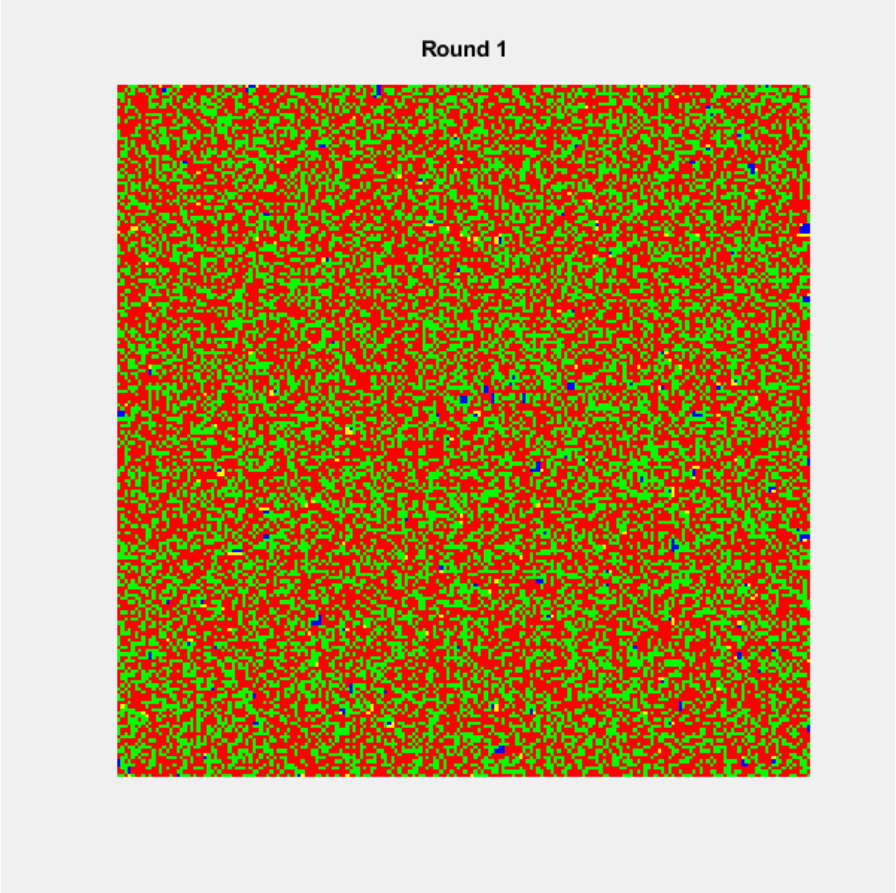


Fig5

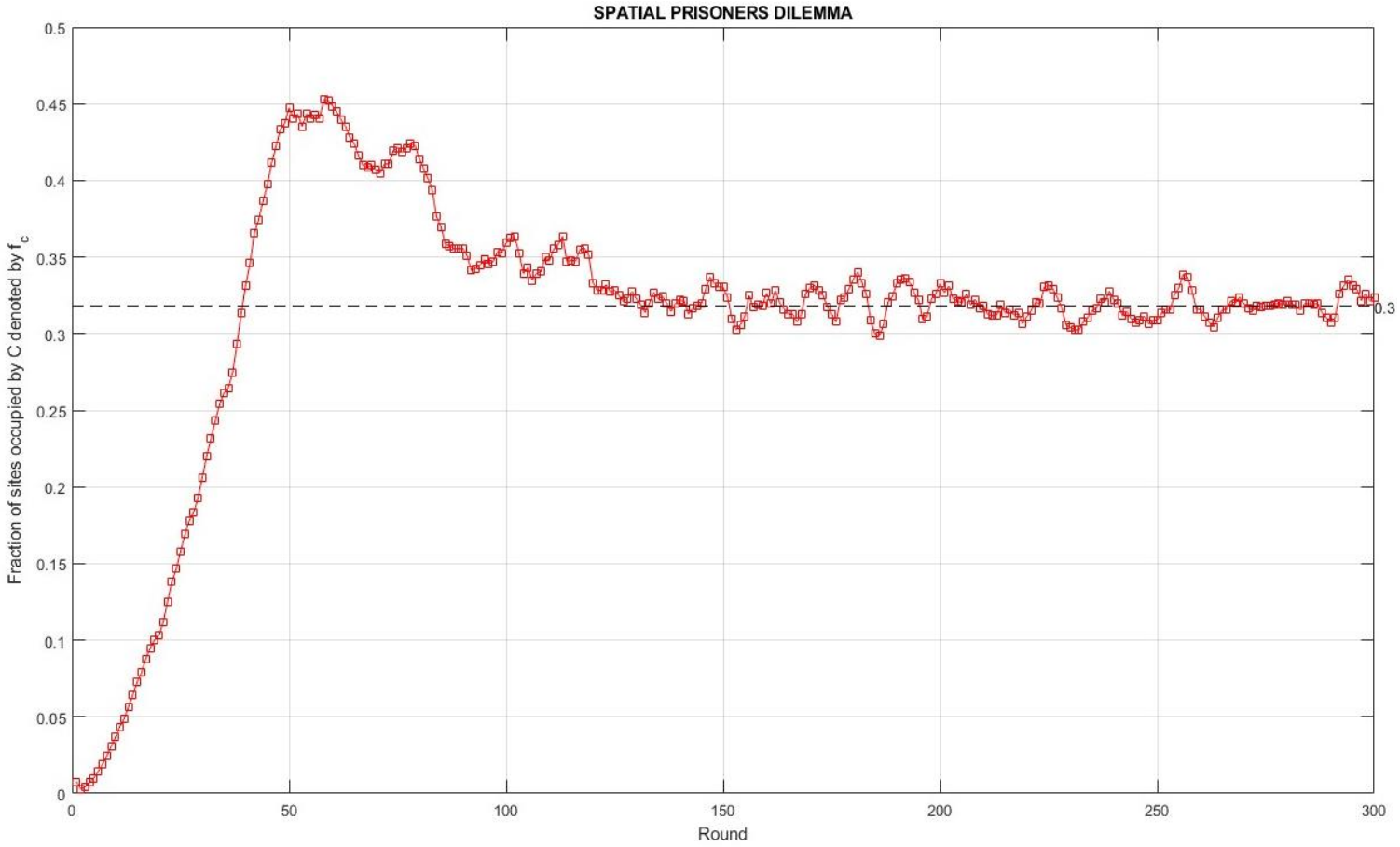


Fig5_fc

Programming simulation---discussion

b value	P value	Self interaction	Fc(final)
1.70	0.1	1	0.72
1.70	0.6	1	0.91
1.90	0.1	1	0.318
1.90	0.6	1	0.318

Programming simulation---discussion

- $b > 1.8$: D clusters grow
- $b < 1.8$: D clusters shrink
- $b < 2$: C clusters can grow
- $b > 2$: C clusters do not grow
- $1.8 < b < 2$: chaotic behavior

Programming simulation

b=1.90;p=0.1;mutation_rate = 0.001;fc≈0.31

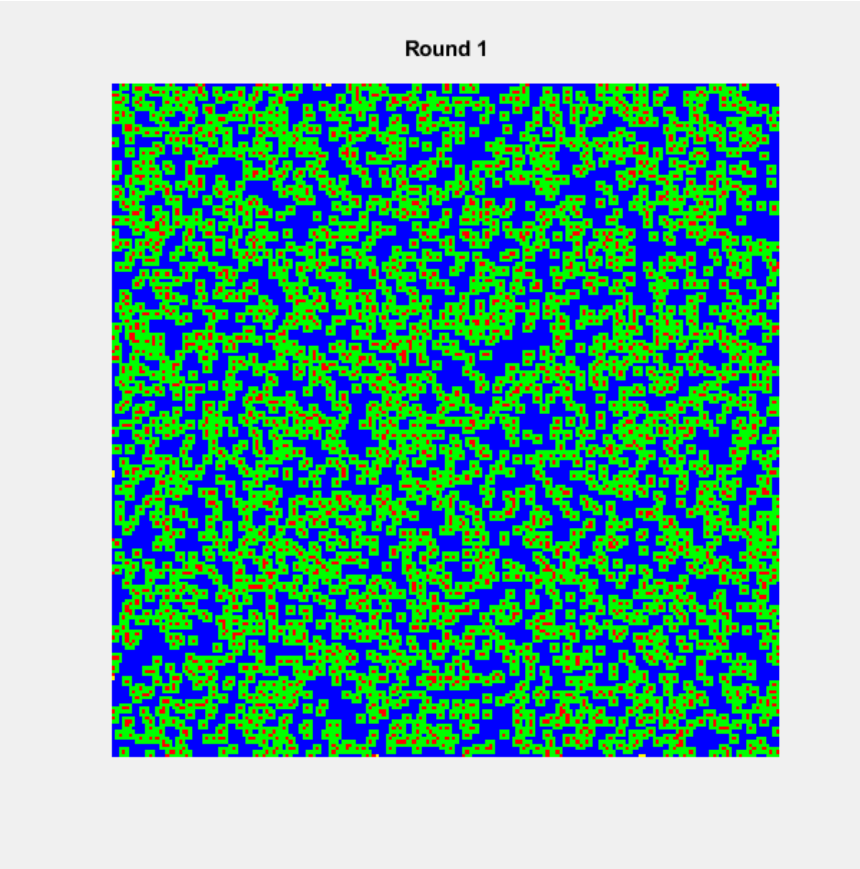


Fig8

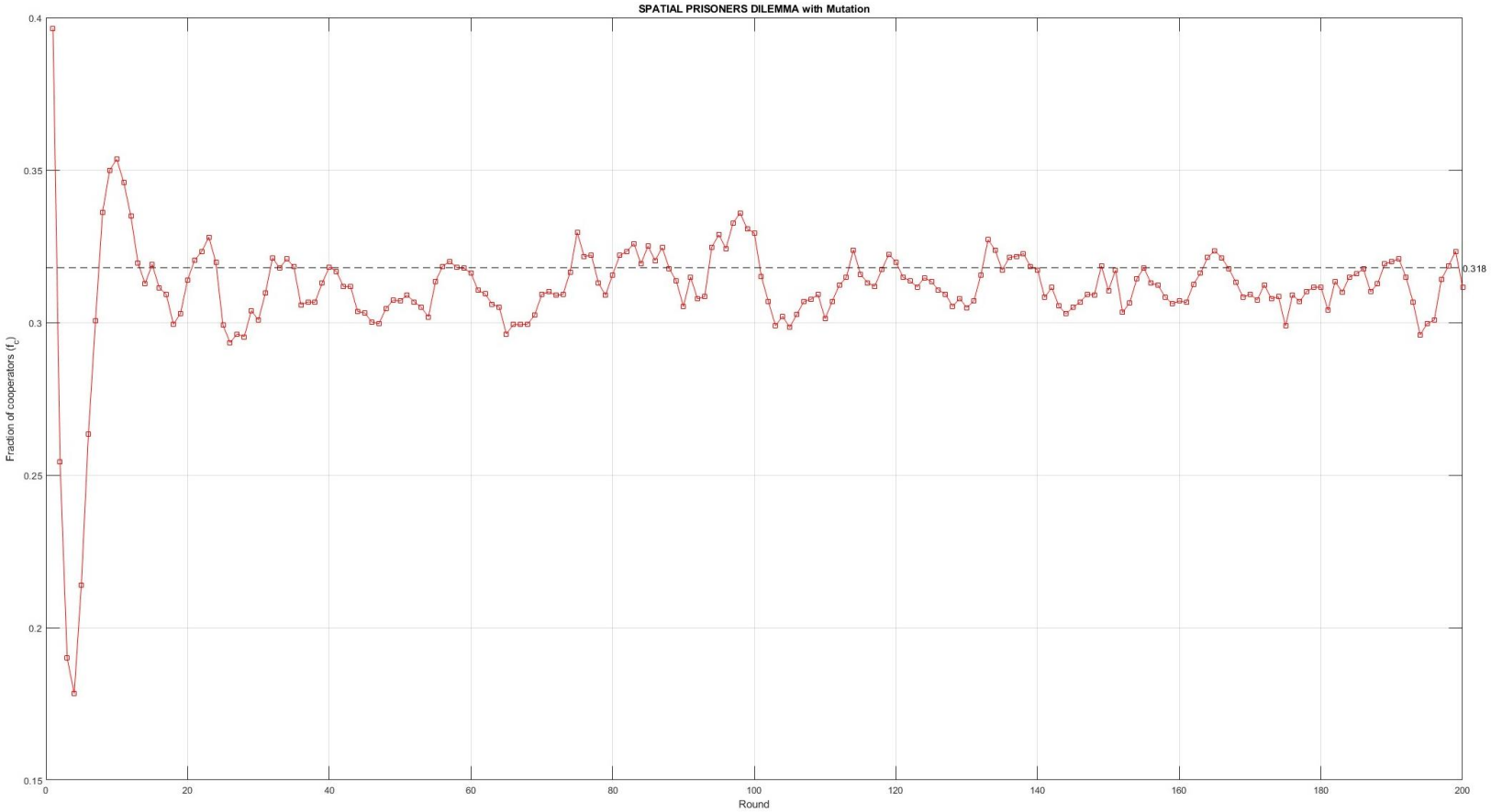


Fig8_fc

Programming simulation

b=1.90;p=0.1;mutation_rate = 0.01;fc≈0.25

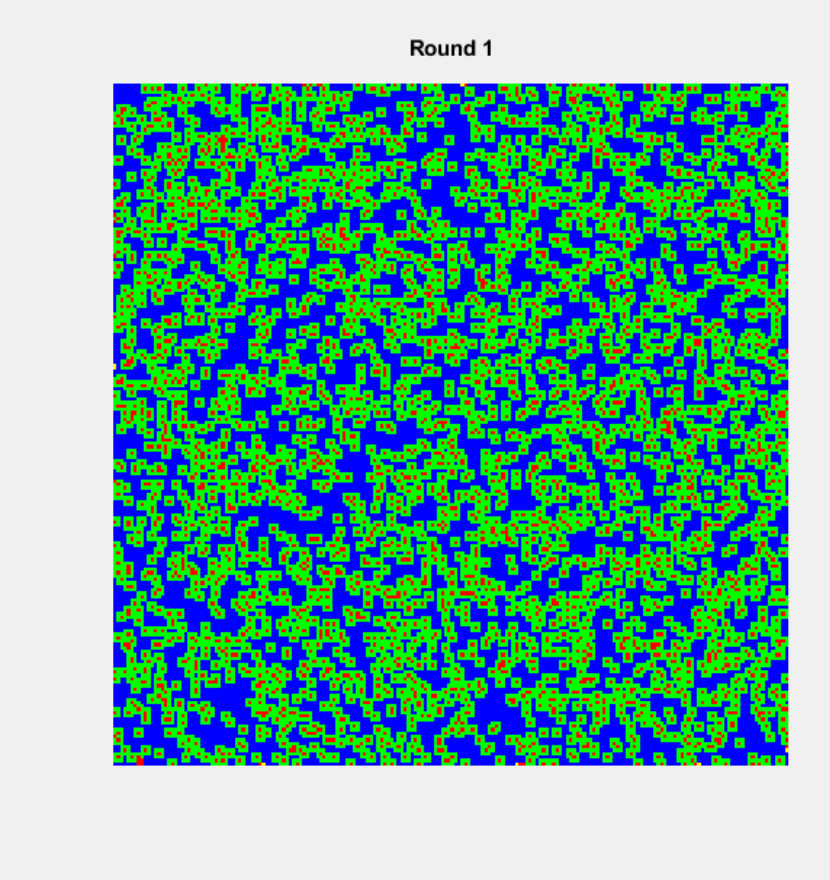


Fig10

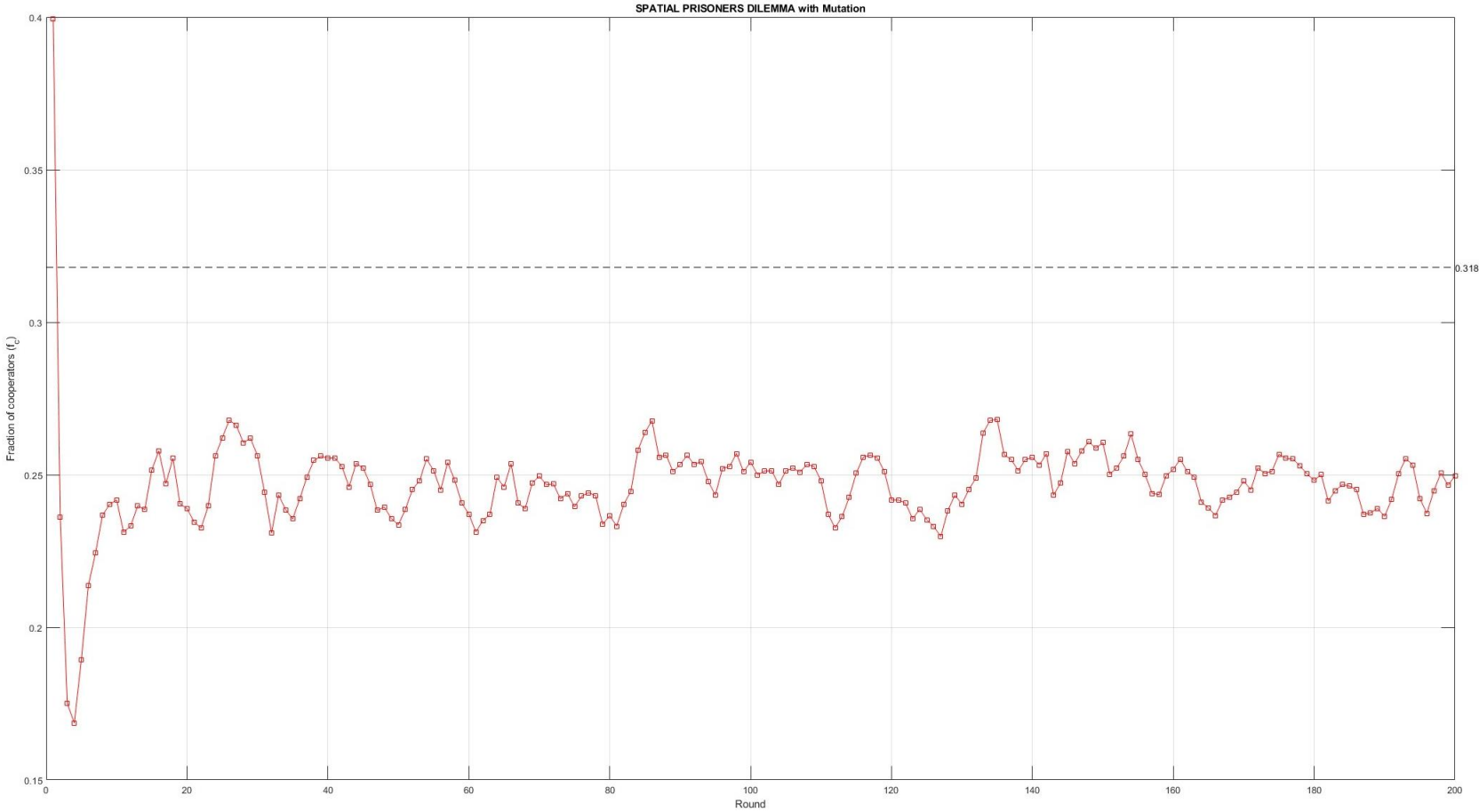


Fig10_fc

Programming simulation

b=1.90;p=0.1;mutation_rate = 0.001;fc≈0.318

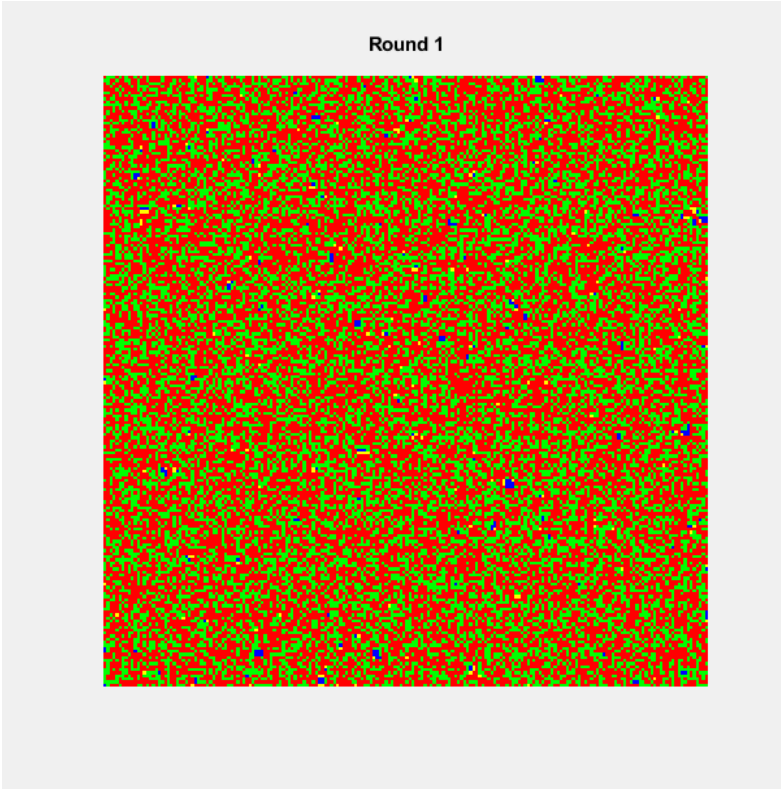


Fig9

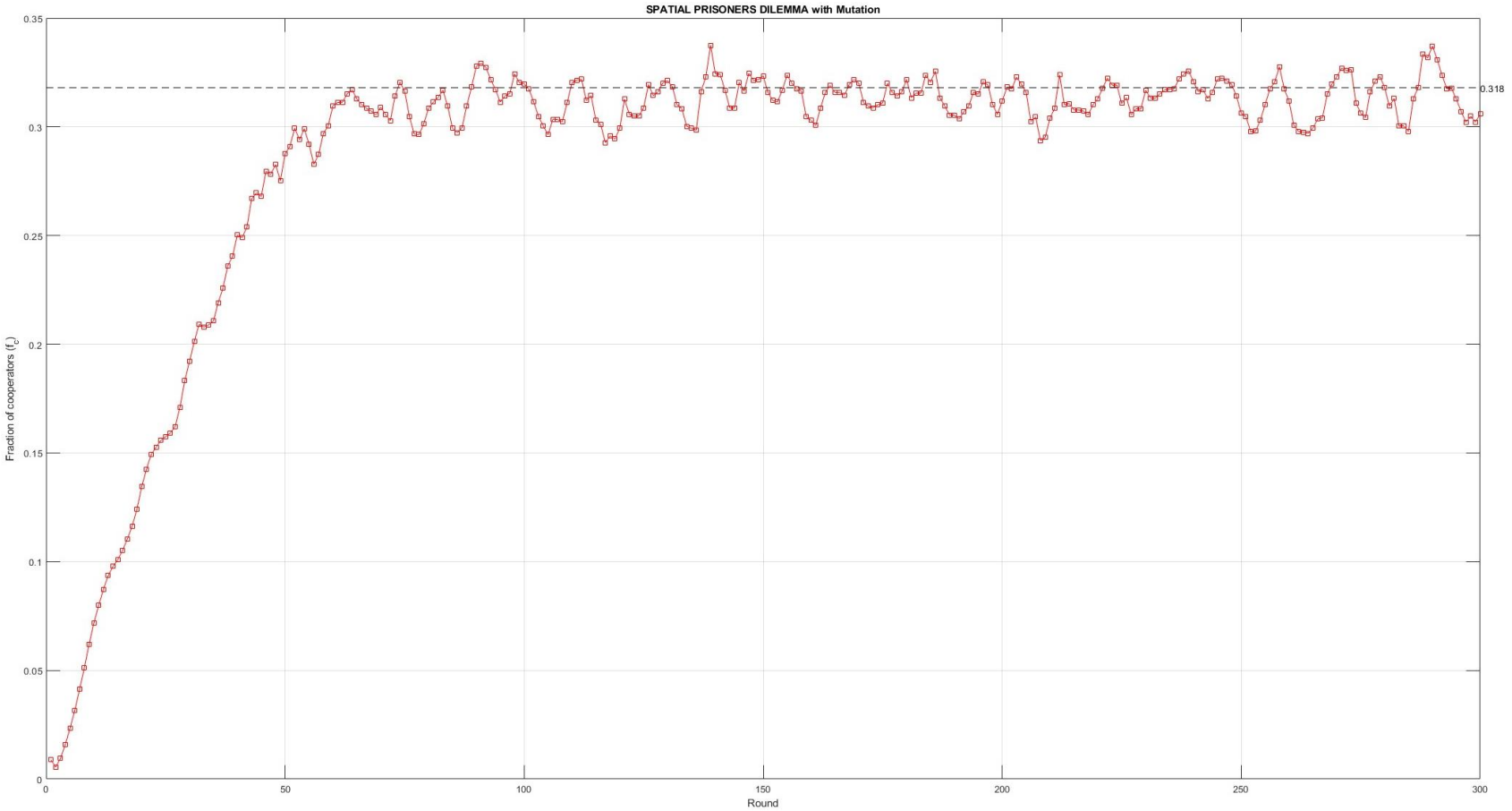
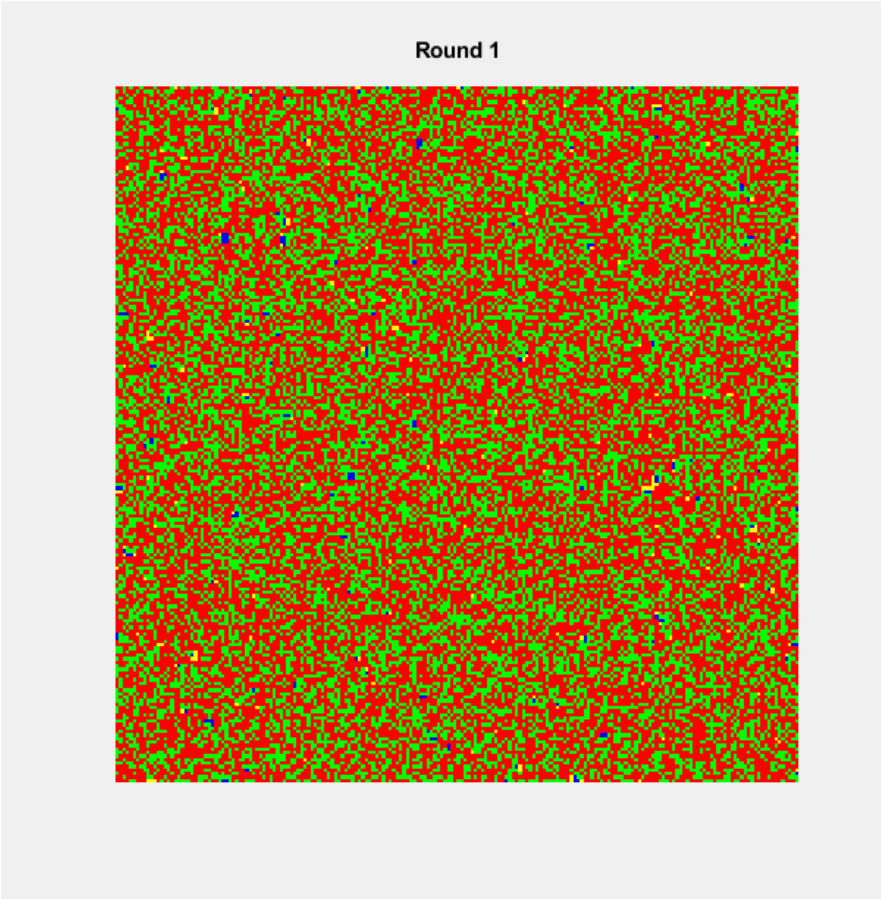


Fig9_fc

Programming simulation

b=1.90;p=0.1;mutation_rate = 0.01;fc≈0.25



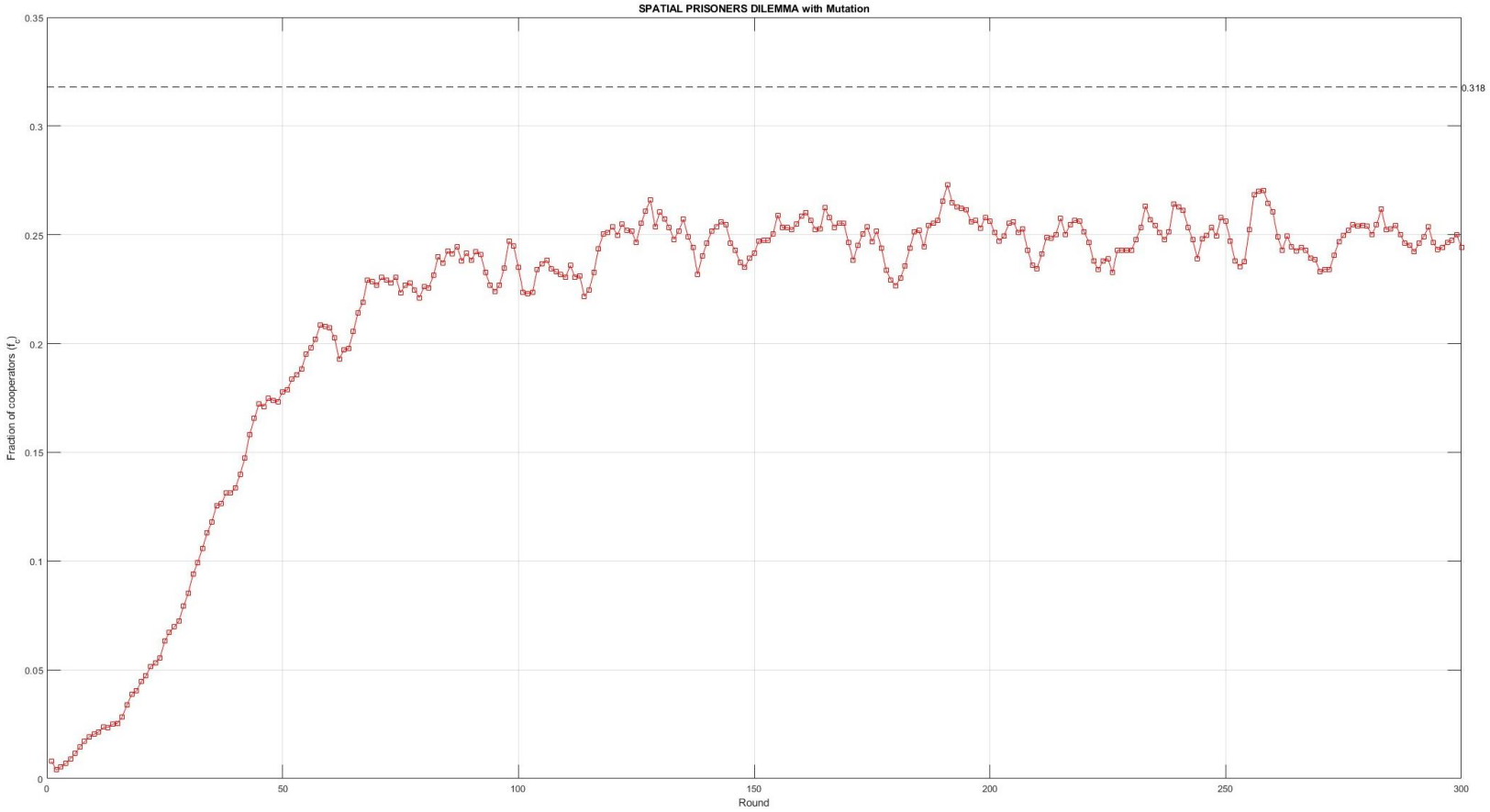


Fig11

Fig11_fc

Programming simulation---discussion

Why does "mutation" decrease the cooperation proportion (fc)?

1. Mutation introduces randomness, disrupting evolutionary trends.

Defectors can mutate into cooperators (aids cooperation).
Cooperators can also mutate into defectors (harms cooperation).

2. The higher the mutation probability, the more volatile the fc fluctuations.

Mutation causes more frequent strategy changes → more frequent pattern changes → making it difficult for cooperative clusters to expand stably.(突變使策略更常變化 → 更常改變格局 → 合作團塊難以穩定擴張)

Mutation Rate突變率	**fc**
Low (e.g., 0.001)	fc decreases slightly
High (e.g., 0.01)	fc decreases significantly

Programming simulation---application

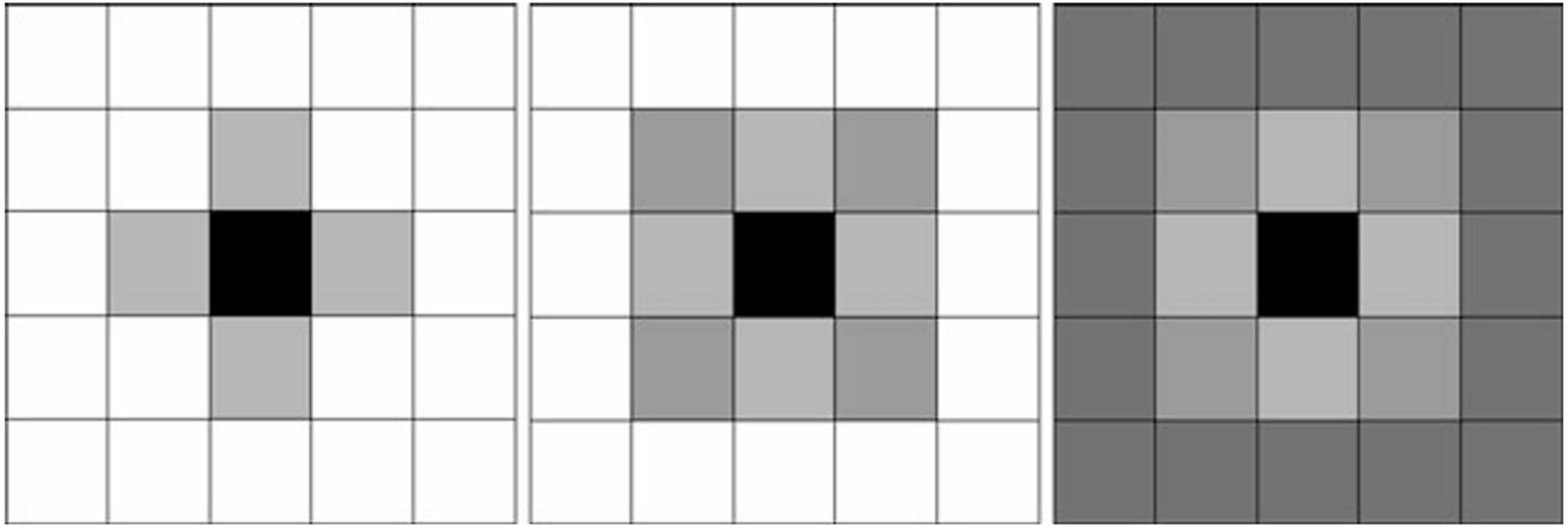
- Origin of molecular cooperation
- Pattern formation in Turing models
- Analogy to 2-state Ising models

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

- Apply our model to the situation where there is a bunch of companies in the same market business competing with each other
- Each company can choose to cooperate with their neighbor or not
- To make this situation more real and more competitive we also decided to introduce a reward policy that gives to the cooperators some advantage

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

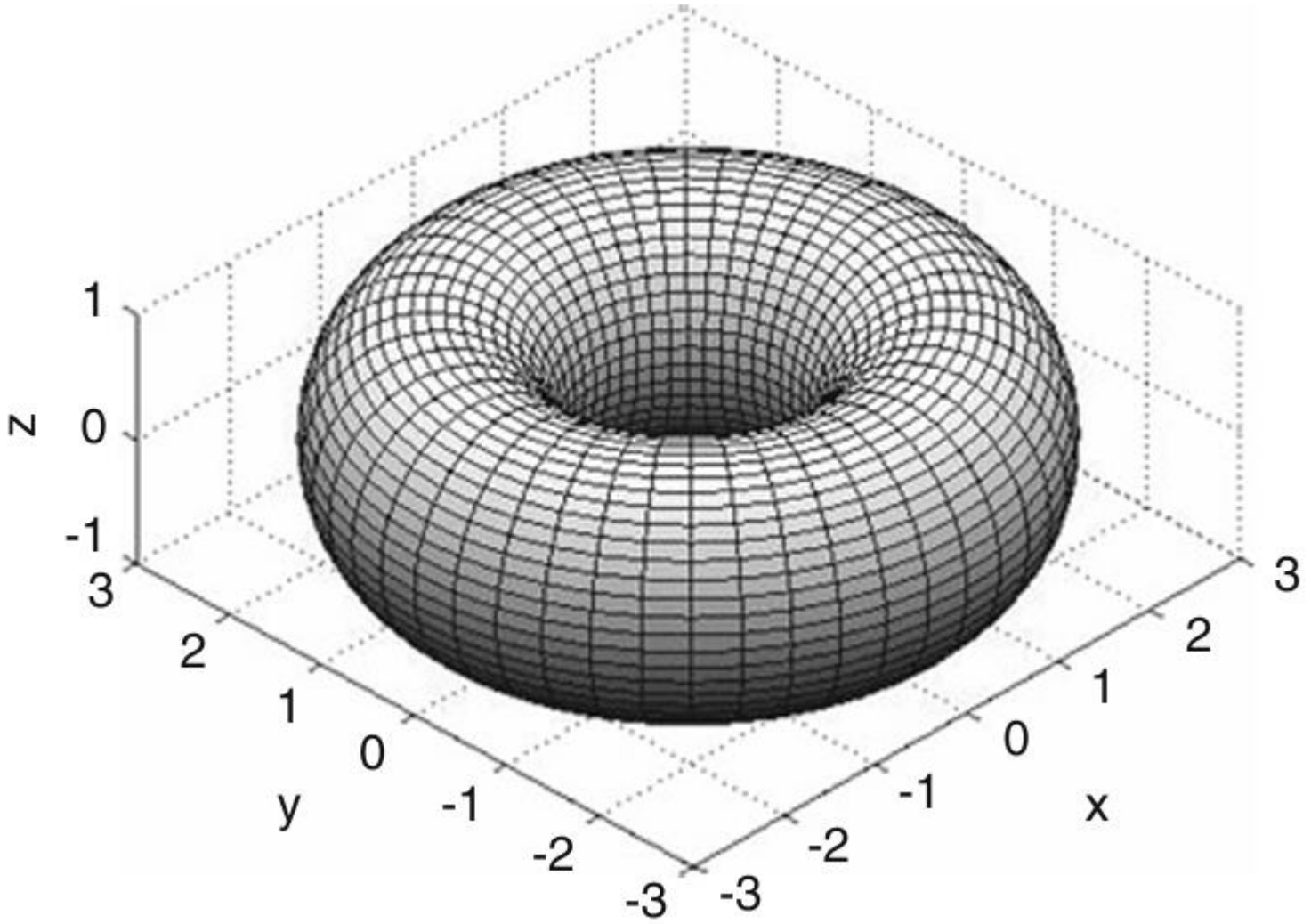
Von Neumann neighborhood	Moore neighborhood	Extended Moore neighborhood



Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

Torus Structure:

By connecting square lattice from left to right side and from up to bottom side we obtain a torus topology, where all cells are equal (all have the same count of neighbors).



Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

Three Behaviors Strategies:

- Greedy --> Copy Best Player
- Conformist --> Copy Best Strategy
- Pavlovian--> Has a learning coefficient that adjusts the probability of cooperation based on past outcomes (reward and punishment adaptation

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

Greedy --> Copy Best Player:

$$M^{i,t} \leftarrow argmax_{j \in N_L^i} \left(\frac{\pi_\epsilon^{j,t}}{n_\epsilon^{i,t}} \right)$$

$$P(x^{i,t+1} = s) \leftarrow \begin{cases} 1 & \text{if } x^{i,t} \in \{x^{j,t} | j \in M^{i,t}\} \text{ and } s = x^{i,t} \\ 0 & \text{if } x^{i,t} \in \{x^{j,t} | j \in M^{i,t}\} \text{ and } s \neq x^{i,t} \\ \frac{\sum_{j \in M^{i,t} \wedge x^{j,t} = s} n_\epsilon^{j,t}}{\sum_{j \in M^{i,t}} n_\epsilon^{j,t}} & \text{otherwise} \end{cases}$$

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

Conformist --> Copy Best Strategy:

$$f_s^{i,t} \leftarrow \begin{cases} \dfrac{\sum_{j \in \cup_s^{i,t}} \pi_\epsilon^{j,t}}{\sum_{j \in \cup_s^{i,t}} n_\epsilon^{j,t}} & \text{if} \sum_{j \in \cup_s^{i,t}} n_\epsilon^{j,t} > 0 \\ -\infty & otherwise \end{cases}$$

$$N^{i,t} \leftarrow argmax_s \left(f_s^{i,t} \right)$$

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

Conformist --> Copy Best Strategy:

$$P(x^{i,t+1} = s) \leftarrow \begin{cases} 1 & \text{if } x^{i,t} \in N^{i,t} \textit{ and } s = x^{i,t} \\ 0 & \text{if } x^{i,t} \in N^{i,t} \textit{ and } s \neq x^{i,t} \\ \dfrac{\sum_{j \in \cup_s^{i,t}} n_\epsilon^{j,t}}{\sum_{\sigma \in N^{i,t}} \sum_{j \in \cup_\sigma^{i,t}} n_\epsilon^{j,t}} & \textit{otherwise} \end{cases}$$

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

Pavlovian(like Win-Stay, Lose-Shift category):

$$RPwt = \sum_{i=1}^{3} Mc_i \cdot w_i$$

$$\alpha_i(t+1) \leftarrow \begin{cases} \alpha_i(t) + 0.15 & \text{, if } (S(t) = S(t-1)) \wedge (S(t-1) = S(t-2)) \\ \alpha_i(t) + 0.10 & \text{, if } (S(t) = S(t-1)) \wedge (S(t-1) \neq S(t-2)) \\ \alpha_i(t) - 0.10 & \text{, if } (S(t) \neq S(t-1)) \end{cases}$$

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

Pavlovian(like Win-Stay, Lose-Shift category):

$$p(t+1) \leftarrow \begin{cases} p(t) + (1 - p(t)) \cdot \alpha_i & \text{, for } S(t) = C \text{ and } RPwt > pf_{avg} \\ (1 - \alpha_i) \cdot p(t) & \text{, for } S(t) = C \text{ and } RPwt \leq pf_{avg} \end{cases}$$

$$q(t+1) \leftarrow \begin{cases} q(t) + (1 - q(t)) \cdot \alpha_i & \text{, for } S(t) = D \text{ and } RPwt > pf_{avg} \\ (1 - \alpha_i) \cdot q(t) & \text{, for } S(t) = D \text{ and } RPwt \leq pf_{avg} \end{cases}$$

$$pf(t) = \frac{\sum C_j}{N} \qquad pf_{avg} = \frac{\sum_i^3 pf_i}{3}$$

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

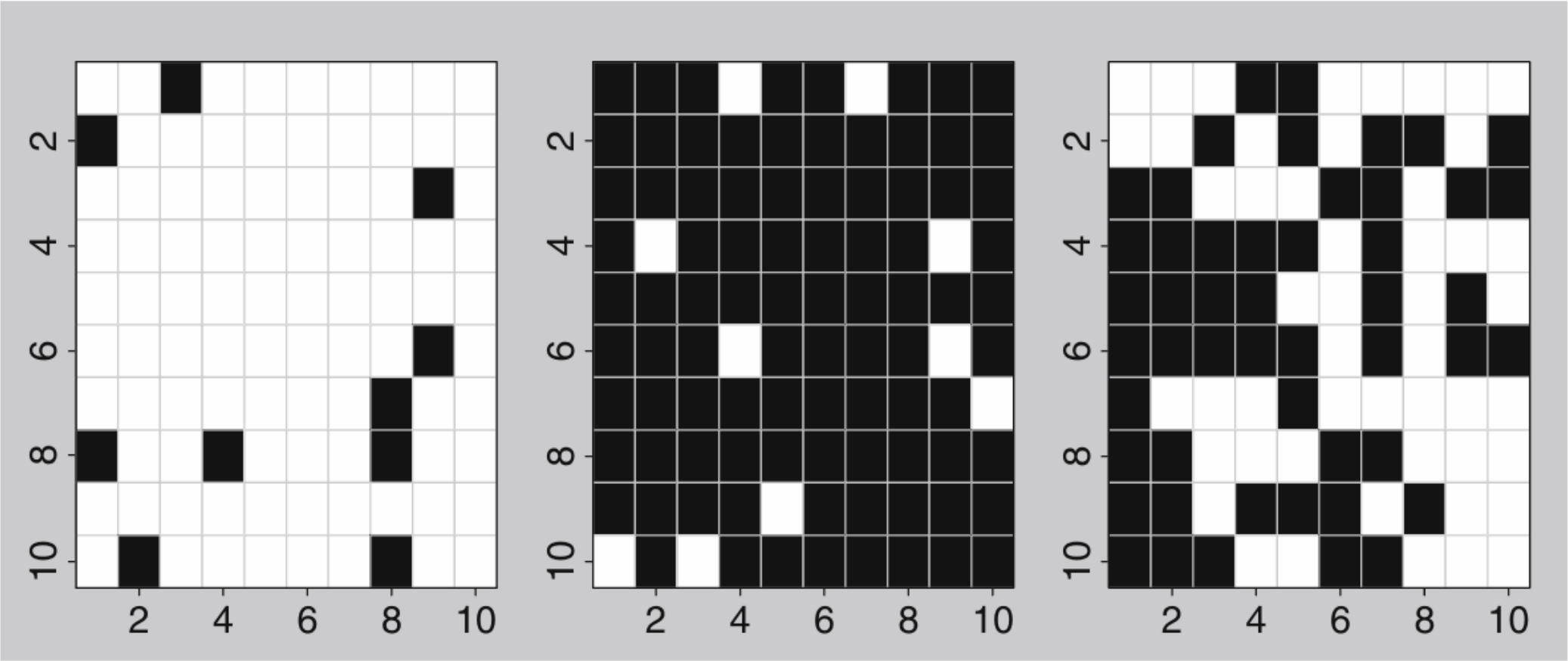
Pavlovian(like Win-Stay, Lose-Shift category):

$$S(t+1) \leftarrow \begin{cases} D & \text{if } RPwt \text{ for agent } i < pf_{avg} \text{ and } p(t+1) < q(t+1) \\ & \text{and } q(t+1) > Ru \\ C & \text{if conditions for } D \text{ are not satisfied} \end{cases}$$

$$S(t+1) \leftarrow \begin{cases} C & \text{if } RPwt \text{ for agent } i < pf_{avg} \text{ and } q(t+1) < p(t+1) \\ & \text{and } p(t+1) > Ru \\ D & \text{if conditions for } C \text{ are not satisfied} \end{cases}$$

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

The initial matrix and condition:



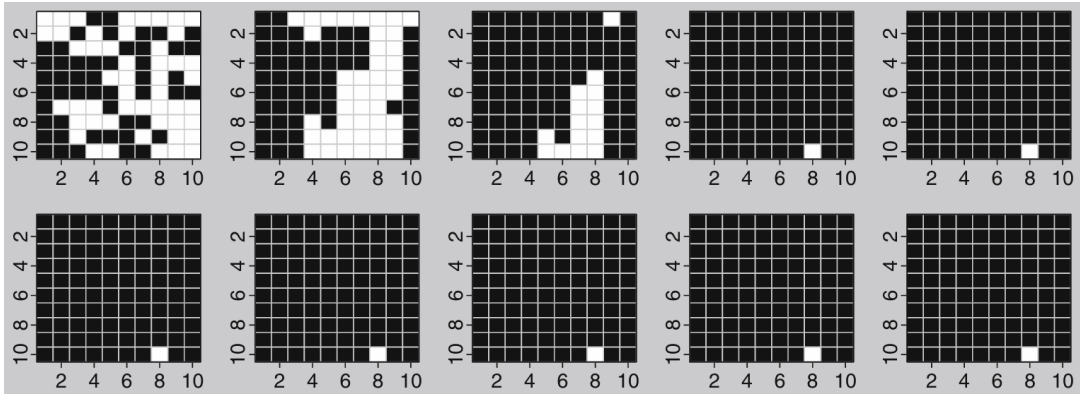
Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

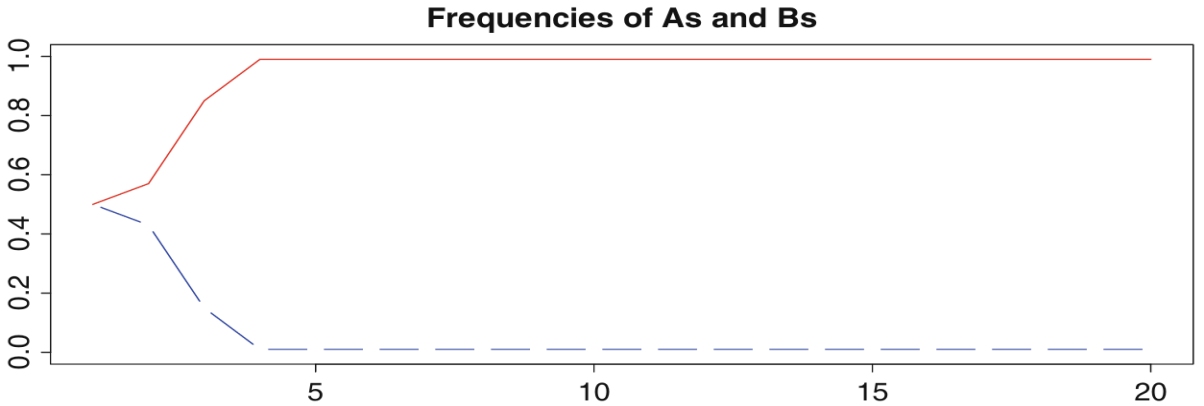
Simulation Setup:

- All agents (companies) adopt the **same behavior type**.
- Three behavioral strategies:
  - **Greedy** (Copy Best Player)
  - **Conformist** (Copy Best Strategy)
  - **Pavlovian** (Win-Stay-Lose-Shift)
- Tested under 3 values of **b** (temptation to defect): **1.1, 1.5, 1.9**

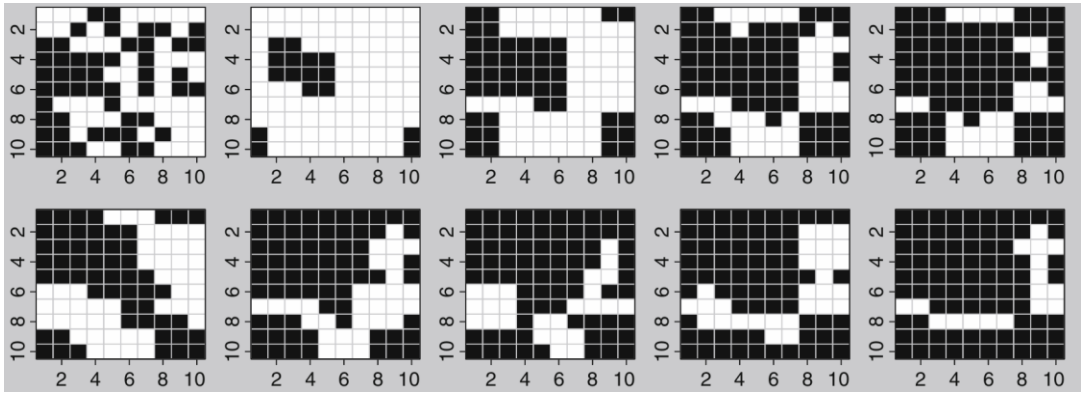
Experiment: Greedy behavior

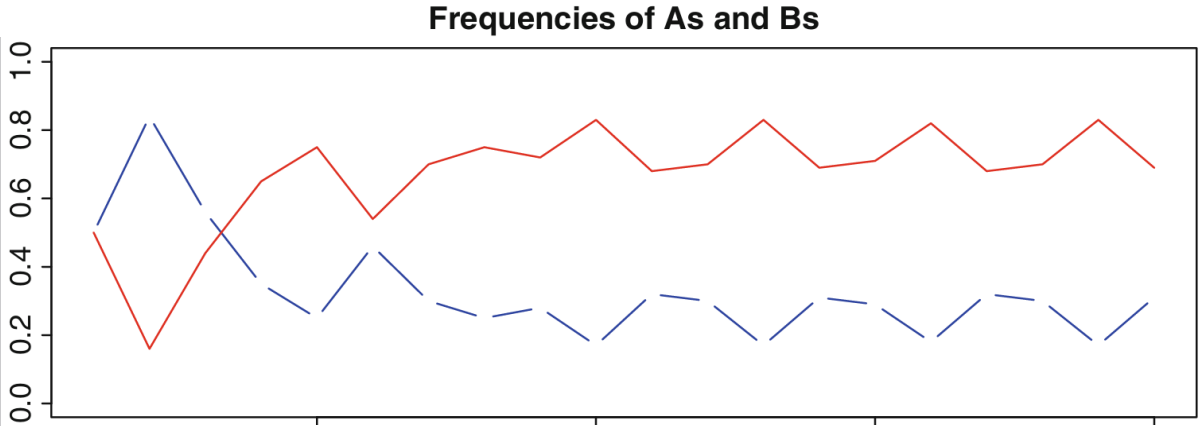
(1)
b = 1.1



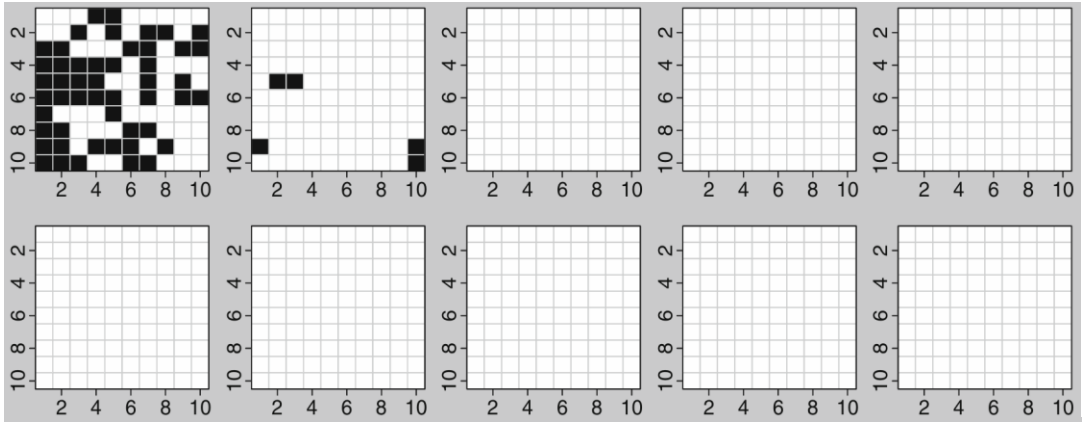


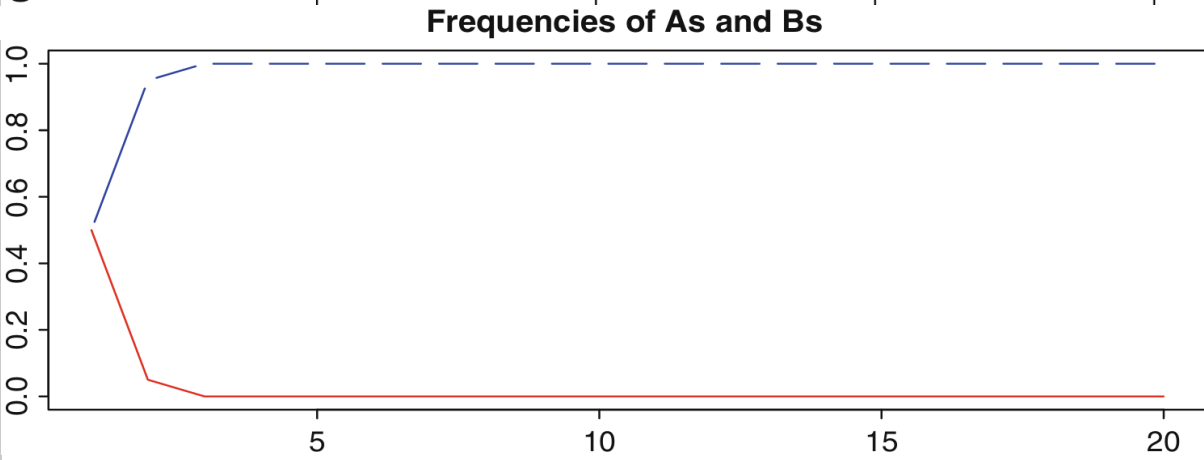
(2)
b = 1.5





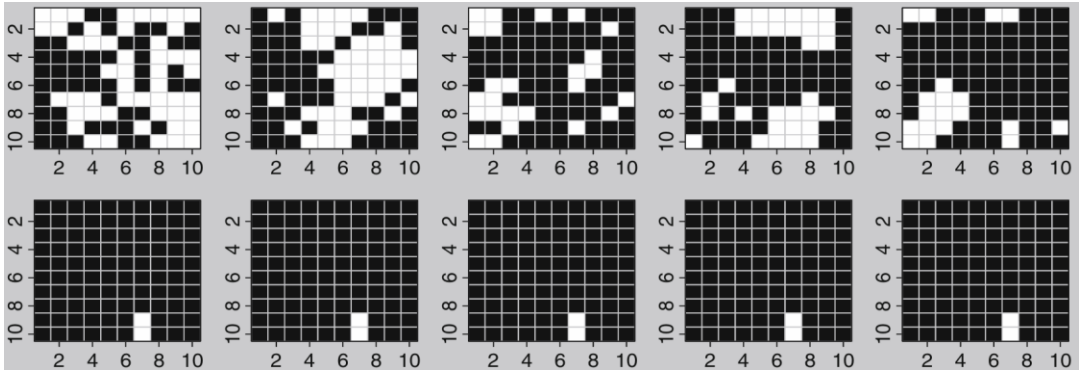
(3)
b = 1.9

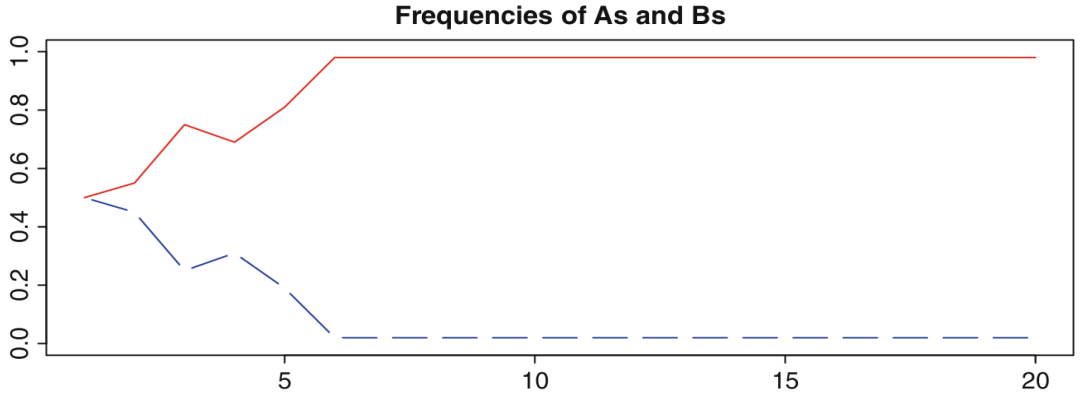




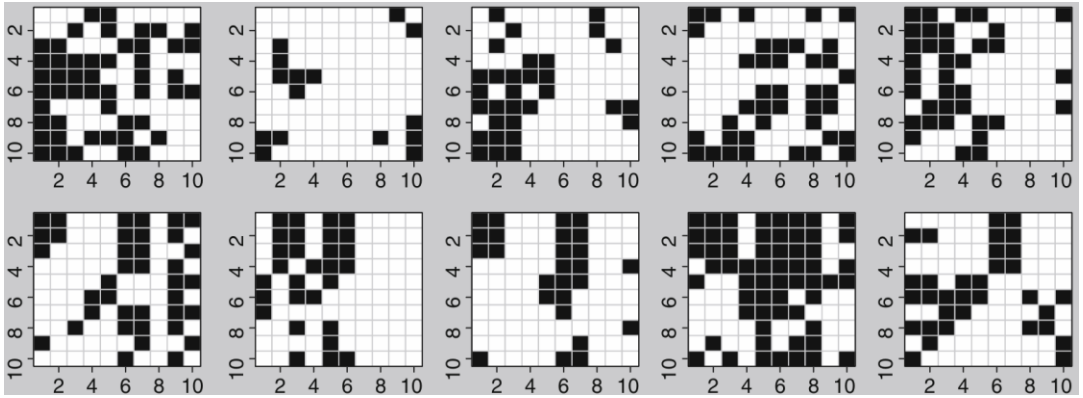
Experiment: Conformist behavior

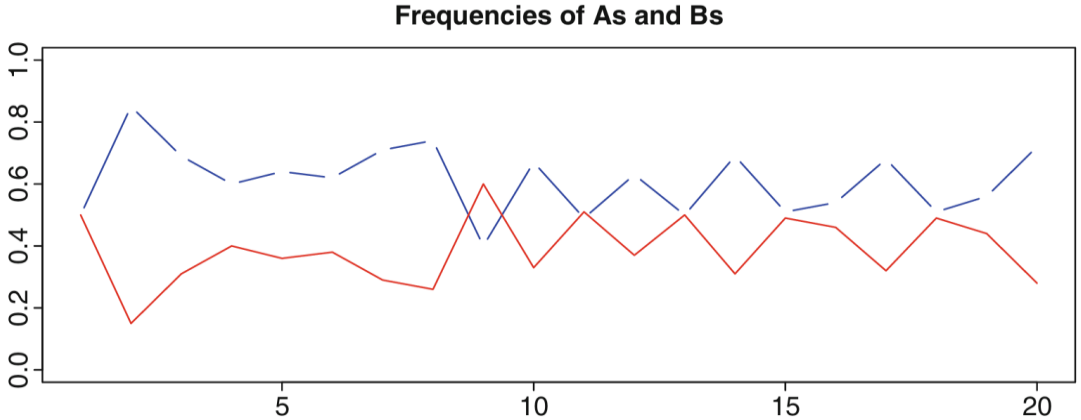
(1)
b = 1.1



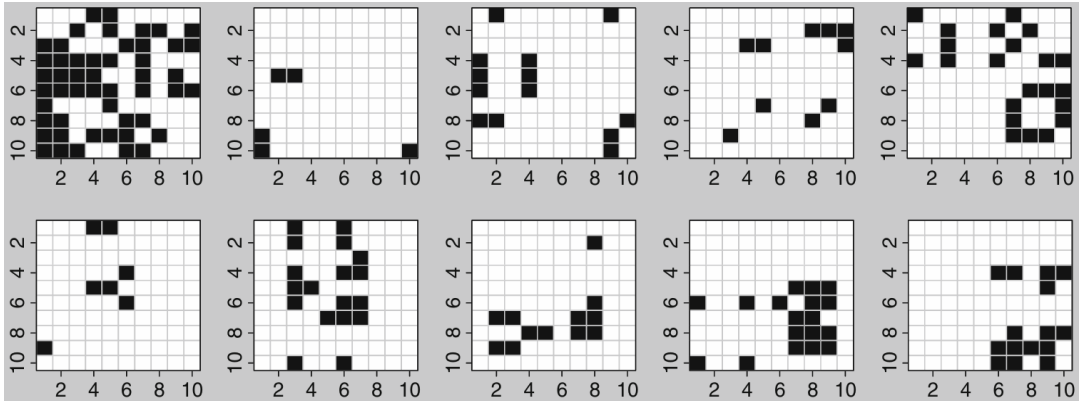


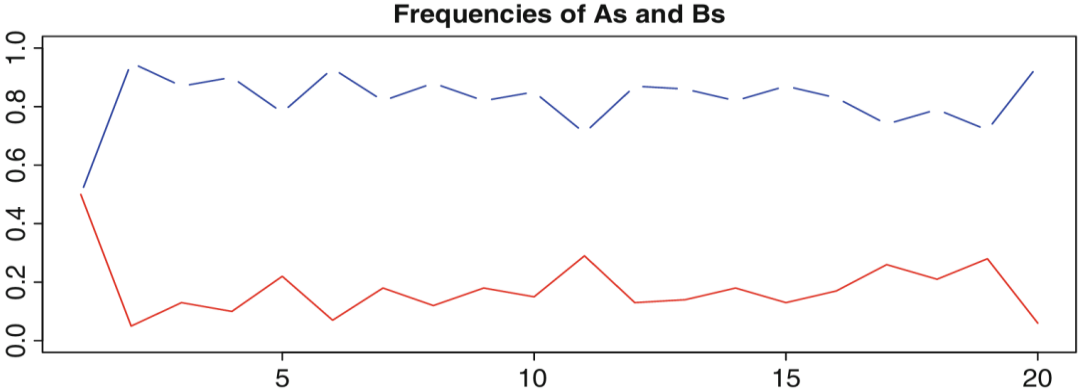
(2)
b = 1.5





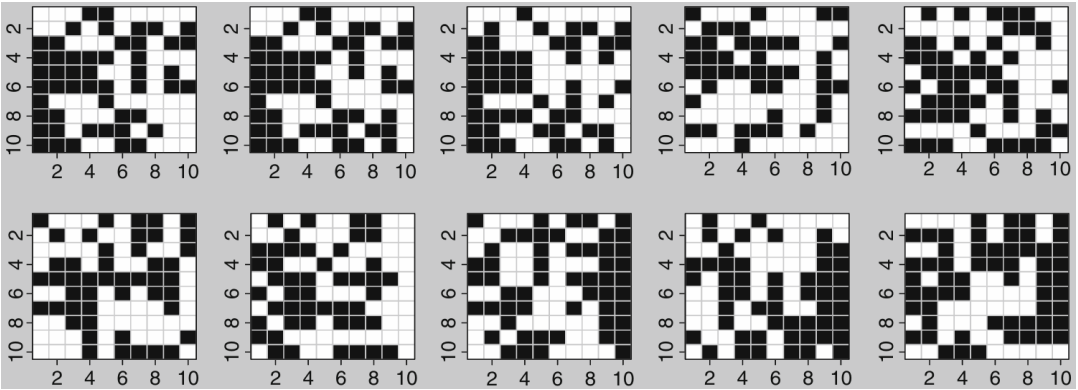
(3)
b = 1.9

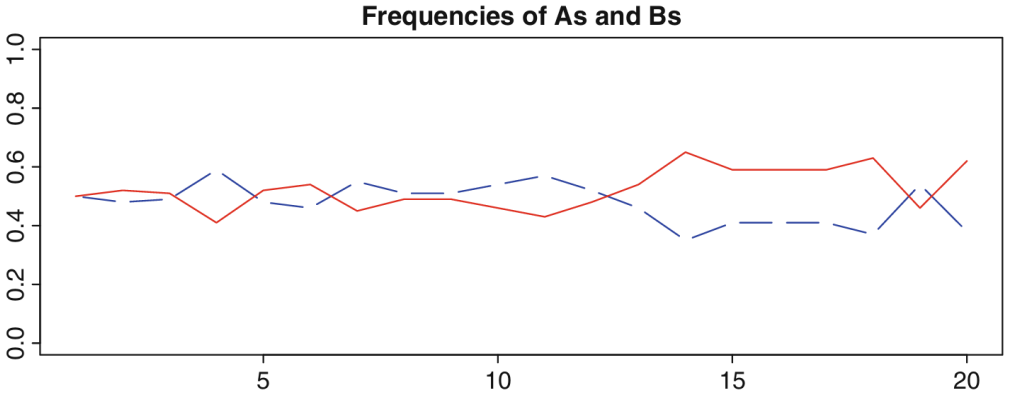




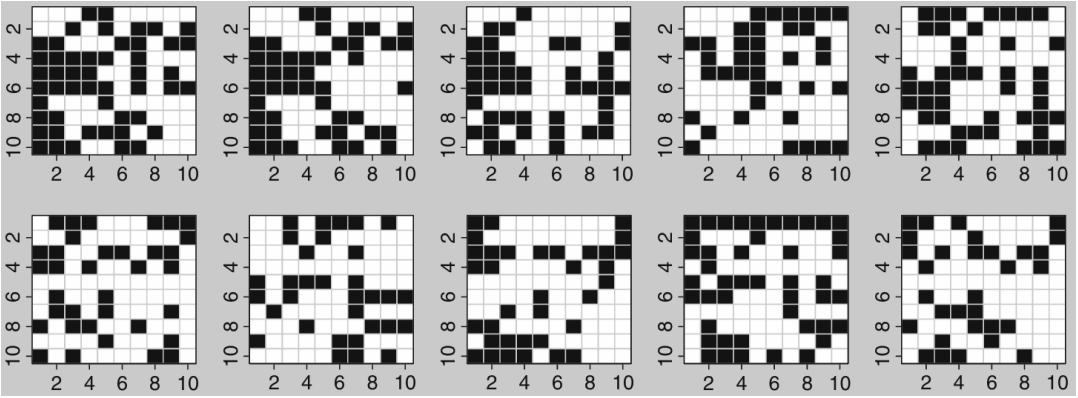
Experiment: Pavlovian behavior

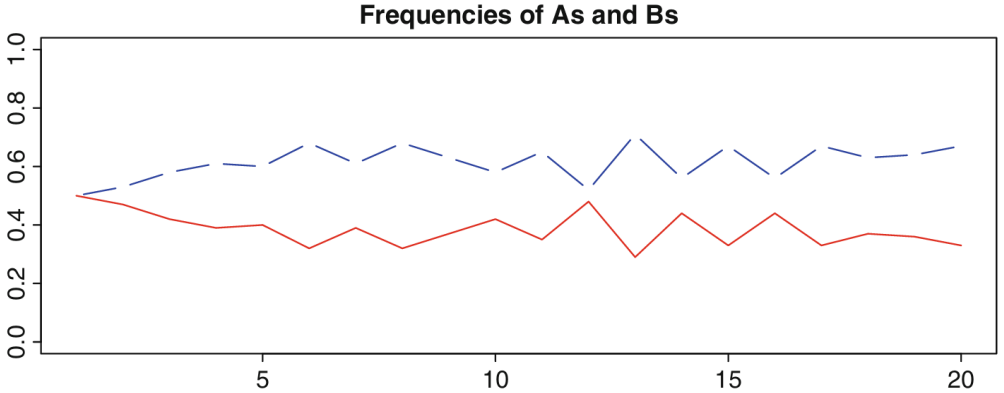
(1)
b = 1.1



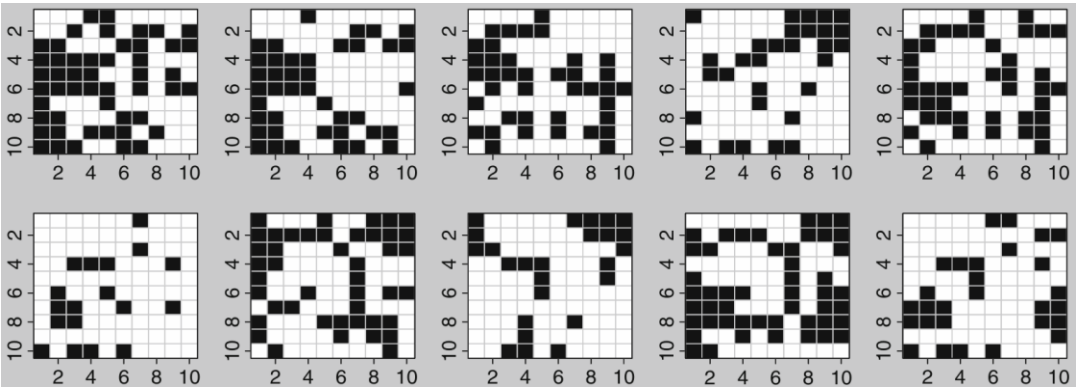


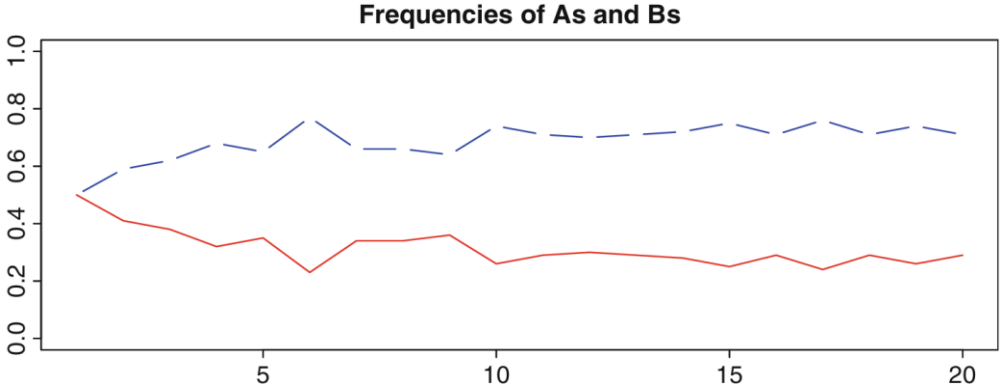
(2)
b = 1.5





(3)
b = 1.9





Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

Mean Payoffs of the three behaviors:

Payoffs(bs)	b = 1.1	b = 1.5	b = 1.9
Greedy	169.3	130.5	8.7
Conformist	162.9	84.8	50.5
Pavlovian	99.9	87.9	90.2

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

Observations

- **b = 1.1**: Cooperation (strategy A) dominates quickly.
- **b = 1.5**: Frequent shifts between strategies; unstable dynamics.
- **b = 1.9**: Defection (strategy B) becomes dominant early.

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

Behavioral Comparison

- **Greedy**: Most sensitive to b changes; poor performance at high b.
- **Conformist**: At medium b (e.g. b=1.5), some cooperators can still be retained, but the stability is not as good as Pavlovian.
- **Pavlovian**: Highest stability at high b; adapts well and maintains high payoff.

Multi-Behavior Scenario

Basic Setup:

- Mix of the 3 behaviors in the population.
- Initial strategy distributions:
  - Scenario 1: 90% Defect, 10% Cooperate
  - Scenario 2: 90% Cooperate, 10% Defect
  - Scenario 3: 50% Cooperate, 50% Defect

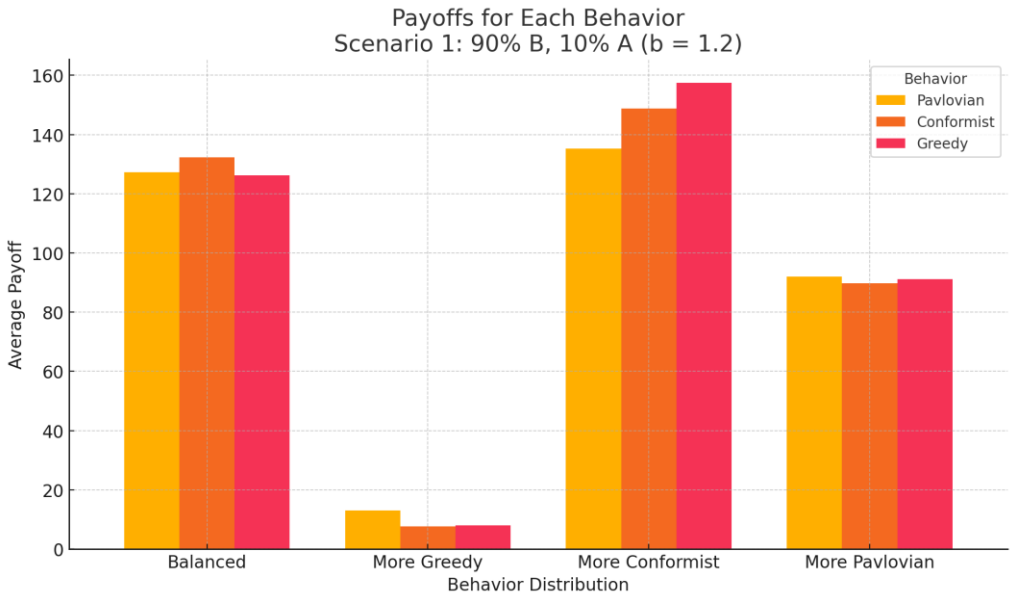
Multi-Behavior Scenario

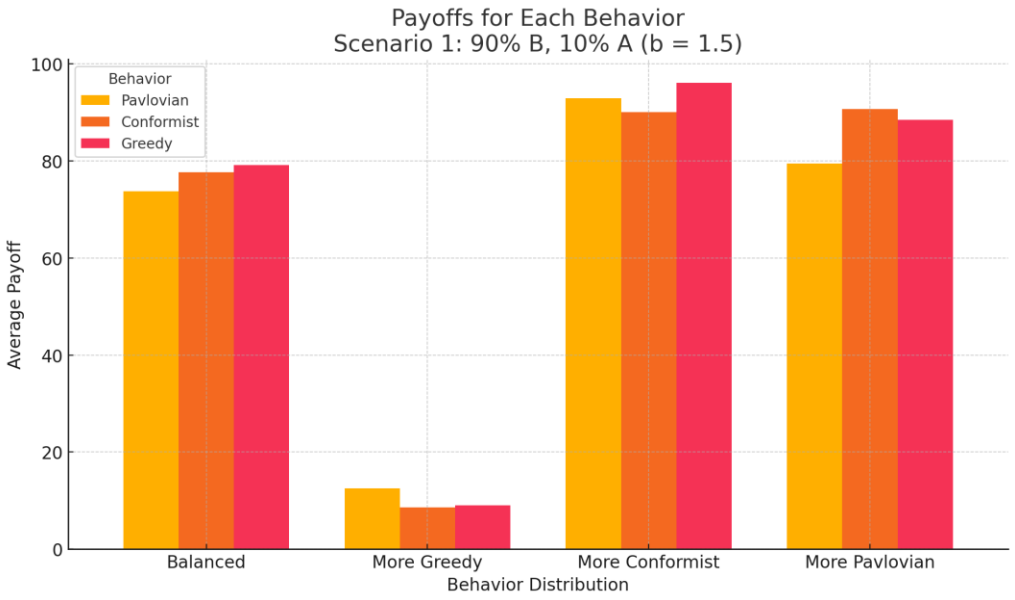
Behavior compositions:

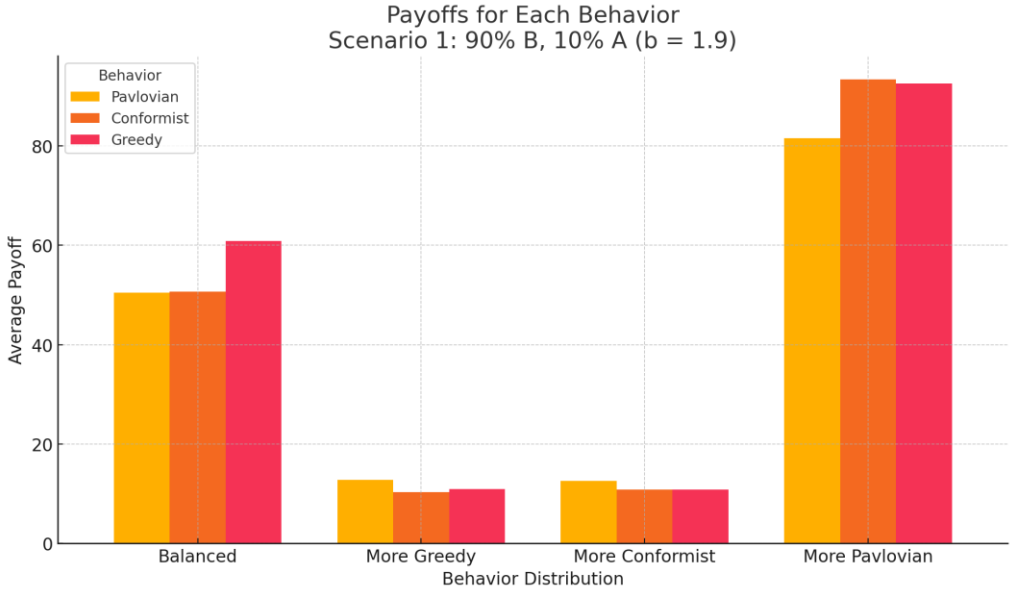
- Balanced (Greedy=33%, Conformist=33%, Pavlovian=34%)
- Greedy-dominant (90% Greedy)
- Conformist-dominant (90% Conformist)
- Pavlovian-dominant (90% Pavlovian)

Multi-Behavior Scenario

Scenario 1:

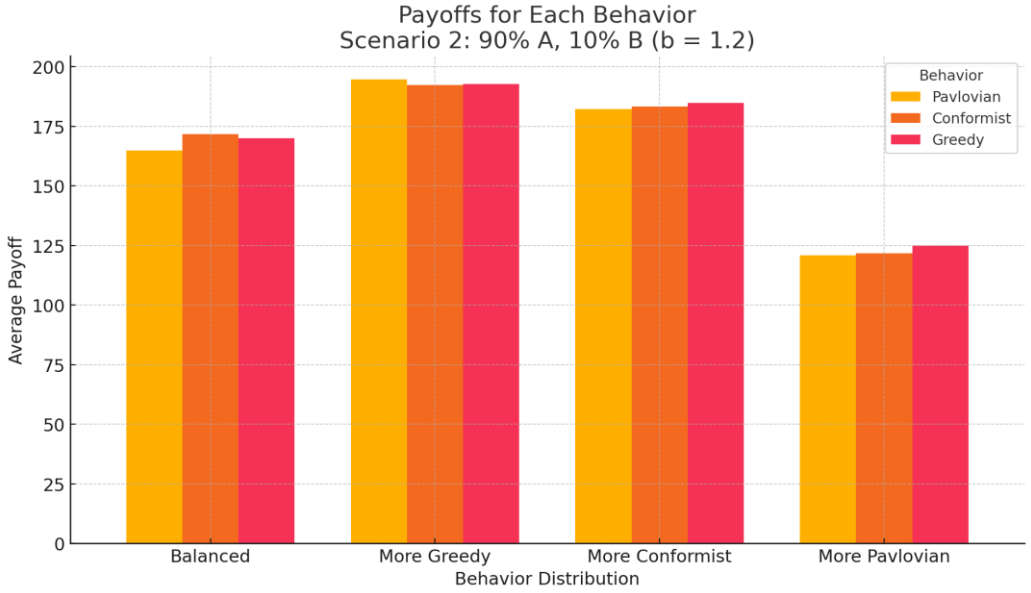


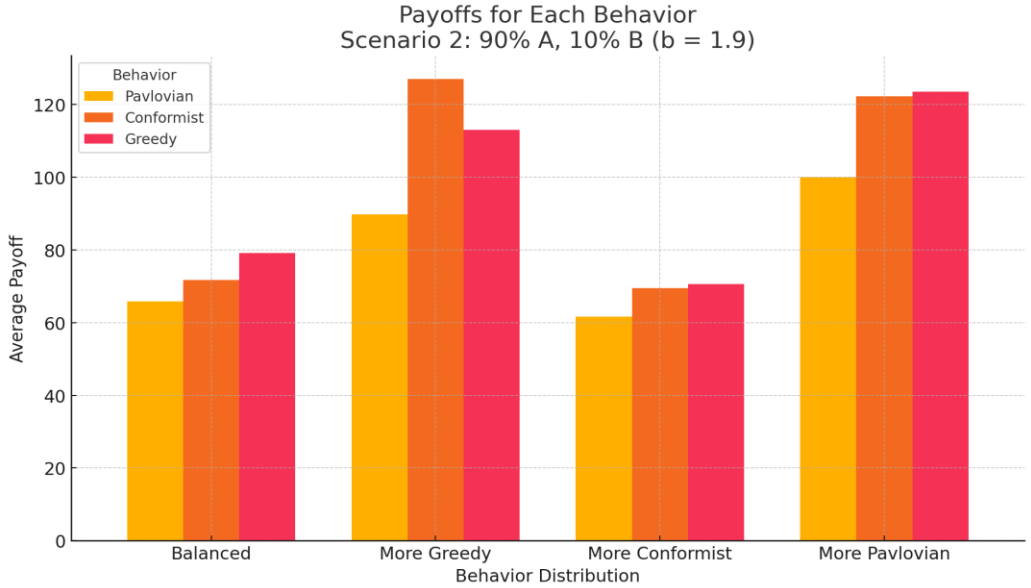


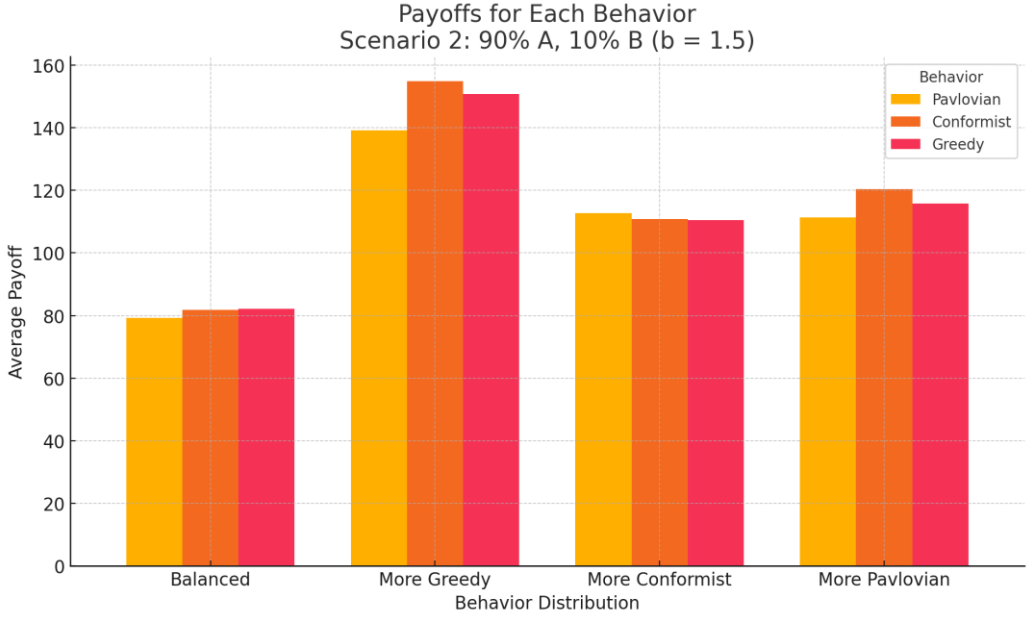


Multi-Behavior Scenario

Scenario 2:

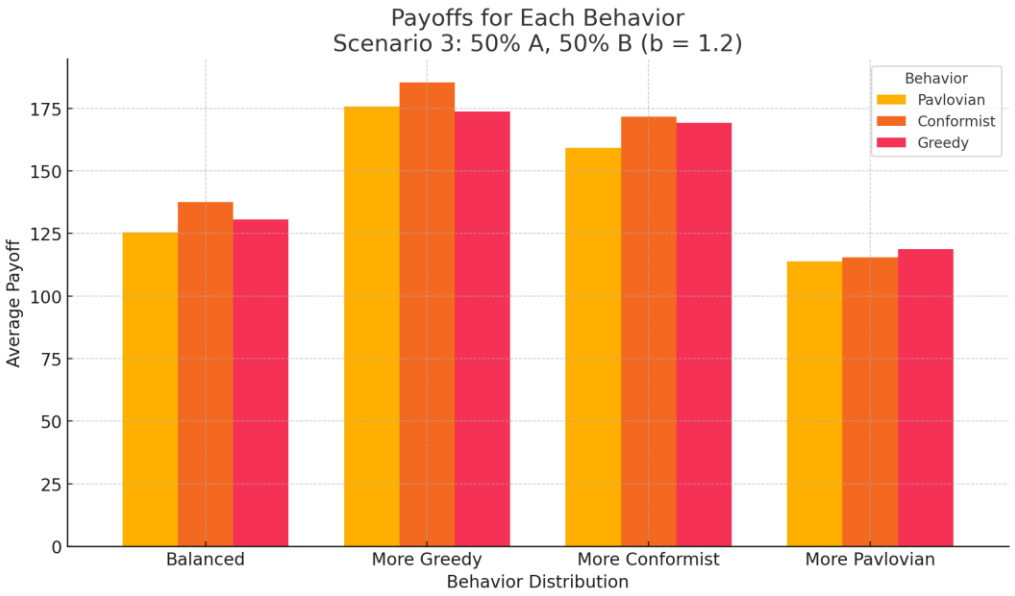


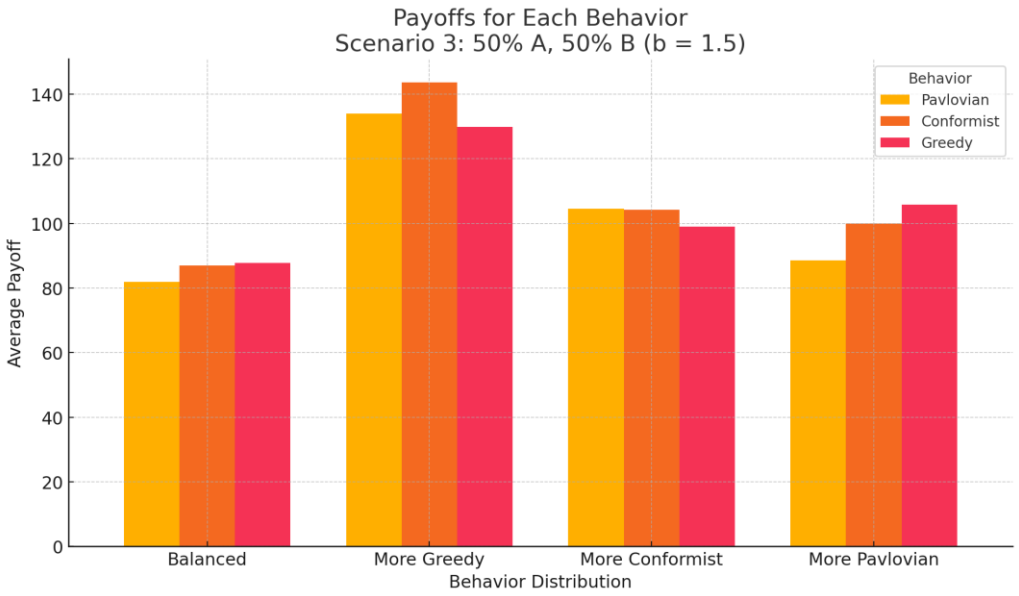


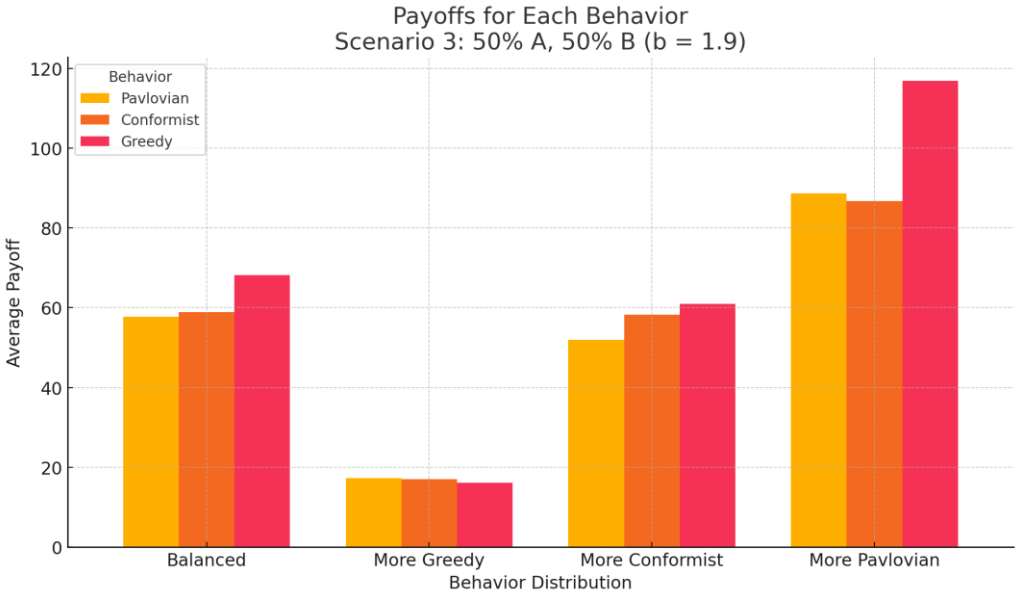


Multi-Behavior Scenario

Scenario 3:







Multi-Behavior Scenario

- **Scenario 1** (high initial defection):
  Pavlovian agents **outperform** others when Greedy dominates.
- **Scenario 2 & 3**:
  No consistent best behavior—**payoffs fluctuate** depending on b and behavior proportions.
- **High b values**:
  Pavlovians **maintain performance** even in high temptation scenarios.

Multi-Behavior Scenario

Link Analysis

- Constructed graphs connecting cooperative agents.
- Graph **density decreases** as b increases:
  - b = 1.1 → 0.0691 (dense)
  - b = 1.5 → 0.0115
  - b = 1.9 → 0.0016 (sparse)
- **Higher cooperation → larger connected clusters**

Multi-Behavior Scenario

Payoff-Based Graph Analysis

- **Lower b values** → agents form **fewer but larger cooperative components**
- **Higher b values** → fragmented network with **many small components**
- Confirmed that **cooperation thrives when b is low**, and link structure reflects this.

Application---An agent-based model for cooperation in the spatial Prisoner's Dilemma

Summary insights

- **Pavlovian agents are the most robust** under high-defection temptation (b = 1.9).
- **Greedy agents** fail in high b settings; overly reactive.
- **Conformist agents** show moderate success, especially in mixed-strategy environments.
- **Social network and graph-based analysis** reveal strong correlation between cooperation and cluster density.

Conclusion

- Basic Dynamics of the Spatial Prisoner’s Dilemma
- Analysis of An agent-based model and Multi-Behavior Scenario

Reference

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- Jaden Cohen & Brandon Holland (2024). *Deep Learning and the Prisoner's Dilemma: A Strategic Evaluation.*
- Nowak, A., & Lewenstein, M. (2014). *An agent-based model for cooperation in the spatial Prisoner's Dilemma*. Journal of Theoretical Biology, 107(2), 321–331.
- *Deep Learning and the Prisoner's Dilemma: A Strategic Evaluation By* Jaden Cohen & Brandon Holland
- Game-Theoretic Deep Reinforcement Learning to Minimize Carbon Emissions and Energy Costs for AI Inference Workloads in Geo-Distributed Data Centers

Reference

- graph of data center power consumption
- Evolutionary Dynamics of the Spatial Prisoner’s Dilemma with Single and Multi-Behaviors: AMulti-Agent Application

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